Gauge-Covariant Gaussian Attention through Lie-Algebra Links

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Abstract

We formulate attention as communication among statistical agents that decide whom to hear and what to take from each sender. Each token holds a Gaussian belief and prior in a private frame, as a section of a statistical bundle associated with a principal $\mathrm{GL}(K)$-bundle over a single context. Each head has four interfaces: the sender emits a key and content, and the receiver forms a query and writes content through a receiving map. Group-valued links $\Theta_{ij} = U_i e^{L_{ij}} U_j^{-1}$ with Lie-algebra generators L_{ij} give the communication graph a lattice gauge structure with inverse edges and holonomy. With square invertible query and key maps, routing is the divergence $D_{\mathrm{KL}}(q_i \| (\Theta_{ij}^{\mathrm{cmp}})_* q_j)$ of the receiver's belief from the sender's belief carried into its coordinates by a group-valued link. With tiled heads and the shared generator of the content maps, the routed multi-head message is a routed oblique projection followed by link transport, and a standard transformer layer is the affine chart with a full residual. The operator is gauge covariant and factors through frame-invariant reference moments. Gaussian-KL routing is exactly a dot product between query expectation parameters and key natural parameters with a key log-partition bias, and reduces to biased dot-product attention under stated conditions. At every positive temperature, routing weights are an exact source posterior and routed content an exact predictive mean. A four-layer exponential-chart model of width $K = 128$ trained on WikiText-103 records test perplexity 34.48 at context 128 in a single seed; per-type tables that give each token a block frame hold most of its parameters. [**Data pending:** report the final-protocol result with matched baselines and seed variability (Table 7).]

Keywords: gauge covariance, Gaussian attention, Lie-algebra links, variational routing, lattice gauge theory, multi-agent communication, information geometry, transformers

1 Introduction

What does a neural language model compute when it predicts the next token, and could a system without a neural architecture compute the same thing? Communication and computation need no neurons: bacteria coordinate gene expression through diffusible signals (Miller and Bassler, 2001), and a slime mold finds shortest paths and builds efficient transport networks (Nakagaki et al., 2000; Tero et al., 2010). Finite chemical reaction networks perform Turing-universal computation with arbitrarily small error probability (Soloveichik et al., 2008), and DNA strand-displacement cascades implement associative memory (Qian et al., 2011). This paper belongs to a program whose working hypothesis is that neural networks are one computational and biological realization of a more general information geometry and dynamics of interacting agents that hold beliefs, compare them, and exchange messages, and whose long-term goal is a language architecture with no neural components.

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This paper builds the first component of that architecture, the communication layer, in exact form and tests it in a language model.

Attention is already a communication protocol: each position issues a query, advertises a key, and offers a value, and routing weights decide which messages each receiver integrates (Vaswani et al., 2017). Graph, mechanistic, and particle accounts describe the same coordination (Gilmer et al., 2017; Joshi, 2025; Elhage et al., 2021; Geshkovski et al., 2023) but leave open what each unit holds, in which coordinates, and how a message written in the sender's coordinates is re-expressed in the receiver's. One geometric construction answers all three.

Each token occurrence is a statistical agent (q_i, p_i, U_i): a Gaussian belief, a prior, and a frame identifying its private coordinates with a shared reference space $\mathbb{R}^K$. The frames are points of a principal $\mathrm{GL}(K)$-bundle and the beliefs are sections of the associated statistical bundle, whose fiber is the Gaussian family $\mathbb{R}^K \times \mathrm{SPD}(K)$ with its Fisher metric (Amari and Nagaoka, 2000) (Section 2.1, Figure 1). The base is a single context c_*, so agents are labeled copies of one fiber differing in frame and belief, and no connection, curvature, or topology of a base enters.

Each head has four maps (Figure 2). With $\xi_j = U_j^{-1}\mu_j$ the mean of sender j in reference coordinates, the sender emits a key $S_K\xi_j$, which makes it findable, and content $T\xi_j$, which is what it says; receiver i forms an untransmitted query $S_Q\xi_i$ and writes received content with a receiving map $\widetilde{O}$, which in the affine chart is the stored factor O. These are the W_K, W_V, W_Q, and W_O of a transformer, a protocol shared by the agents of a layer. The routing weights β_{ij}, the decision of whom to hear, are a Gibbs distribution over relative entropies between the receiver's query law and the senders' key laws; for square invertible query and key maps the comparison is

$$D_{\mathrm{KL}}\big(q_i \| (\Theta_{ij}^{\mathrm{cmp}})_* q_j\big), \qquad \Theta_{ij}^{\mathrm{cmp}} = U_i S_Q^{-1} S_K U_j^{-1}, \tag{1}$$

the divergence of the receiver's belief from the sender's, carried into the receiver's coordinates by a group-valued comparison link that is the flat identification $\Omega_{ij} = U_i U_j^{-1}$ when $S_Q = S_K$ (Proposition 1).

Content is carried by group-valued links $\Theta_{ij} = U_i e^{L_{ij}} U_j^{-1}$ with Lie-algebra generators $L_{ij} \in \mathfrak{gl}(K, \mathbb{R})$ in reference coordinates. They are invertible and obey the endpoint law $\Theta_{ij} \mapsto g_i \Theta_{ij} g_j^{-1}$ of lattice link variables under frame changes $U_i \mapsto g_i U_i$ (Wilson, 1974; Creutz, 1983), giving the communication graph a lattice gauge structure: a lattice of agents above one point, not a discretization of a base. Its lattice-gauge data are the invertible links, which compose, invert, and carry holonomy; the map $U_i \widetilde{O}_h T_h U_j^{-1}$ of a single head has rank at most the head width and is sheaf-type data on the same graph (Section 3.1). The implemented model shares one generator $L = OT$ across the edges of a layer, so its links are composites of site data; independent edge generators governed by an action are treated in Supplement S6.

For this shared generator the group link splits exactly into the flat transport Ω_{ij} and a selective channel term that vanishes on the directions the heads do not read, takes values in those they write, and has the head maps of multi-head attention as its first-order part (Proposition 6). With tiled heads (widths summing to K) and routing outside the exponent,

the routed message is, in every parametrization,

$$m_i = \sum_j U_i f(L) \Pi(\boldsymbol{\beta}_{ij}) U_j^{-1} \mu_j, \qquad \Pi(\boldsymbol{\beta}_{ij}) = \sum_h \beta_{ij}^{(h)} \Pi_h, \tag{2}$$

with complementary oblique projectors Π_h fixed by the sending maps T_h and $f(L)$ equal to L, $I + L$, or e^L in the affine, first-order, and exponential charts (Theorem 7). Multi-head attention is thus a routed oblique projection followed by link transport. With one head the exponential chart gives the lattice hopping term $\sum_j \beta_{ij} \Theta_{ij} \mu_j$, and a standard transformer layer is the affine chart with a full residual (Section 3).

The operator is covariant under independent frame changes and factors through the frame-invariant reference moments ξ_i and $C_i = U_i^{-1} \Sigma_i U_i^{-\top}$ (Theorem 11), so covariance constrains none of the shared maps: the frames are a dressing field (Fournel et al., 2014; François, 2019), not a symmetry that forces intertwiners, and act in reference coordinates as multiplicative, token- and position-conditioned linear maps. Gaussian-KL routing is exactly a dot product between query expectation parameters and key natural parameters, offset by the key log-partition function (Theorem 13), and with a shared projected key covariance it reduces to dot-product attention in whitened coordinates with a key-norm bias (Proposition 14). At every positive temperature the routing weights are an exact source posterior and routed content an exact predictive mean (Theorem 15).

A four-layer exponential-chart language model of width $K = 128$ with four heads of width 32, in which normalization, a multilayer perceptron (MLP), and a linear decoder surround the communication layer, records token-level test perplexity 34.48 on WikiText-103 at context 128 in a single seed. Here K is the width of the evolving state, not a d_{model}-equivalent capacity: per-type tables that give each vocabulary entry a $K \times K$ block-diagonal frame besides its prior mean and variance hold about 219 million of the 227 million parameters, and the conventional baselines matched in non-embedding parameters to the affine run at this width differ there and in the positional frames (Section 7.3).

The prototype retains normalization, the MLP, the decoder, and backpropagation, so it is not the program's non-neural architecture, and claims no free energy for the whole stack. Its links share one generator per layer, and its comparison with transformers is matched neither in protocol nor in per-type parameters.

Sections 2 through 6 develop the theory, Section 7 reports the language model, Section 8 discusses limits and the non-neural path, and the supplement holds secondary material.

1.1 Related work

Attention as communication. Learned communication among agents (Foerster et al., 2016; Sukhbaatar et al., 2016) became attention based with targeted and gated messaging (Jiang and Lu, 2018; Das et al., 2019), and agents trained on shared tasks develop symbol systems (Lazaridou et al., 2017; Mordatch and Abbeel, 2018; Lazaridou and Baroni, 2020). In TarMAC, the closest of these, a message is used in the receiver's coordinates exactly as sent; here every message is transported between frames.

Probabilistic attention. Attention weights have been derived as alignment posteriors (Kim et al., 2017; Deng et al., 2018; Bahuleyan et al., 2018; Fan et al., 2020), as Gaussian-mixture responsibilities with the key-norm term that L2 self-attention also retains (Movellan

and Gabbur, 2020; Gabbur et al., 2021; Nguyen et al., 2022; Kim et al., 2021), and from regularized, kernel, and energy formulations (Martins and Astudillo, 2016; Niculae and Blondel, 2017; Tsai et al., 2019; Ramsauer et al., 2021; Hoover et al., 2023). Gaussian word embeddings are compared by relative entropy (Vilnis and McCallum, 2015; Athiwaratkun and Wilson, 2017), as are token-specific distributions in Pro-Attention (Bae et al., 2022). The routing here recovers the mixture form for a point query and shared key covariance, while a full query covariance adds a pair-dependent term and the compared laws sit in different learned frames.

Group structure in attention. Rotary encodings rotate queries and keys by position (Su et al., 2024), LieRE learns the rotation generators (Ostmeier et al., 2025), and geometric transform attention applies $\rho(g_i g_j^{-1})$ built from known poses (Miyato et al., 2024); each is a flat link Ω_{ij} with given frames and zero generator. Gauge-equivariant networks (Cohen et al., 2019; Weiler et al., 2021; He et al., 2021; Fuchs et al., 2020; Bronstein et al., 2021) treat the frame as geometric data defined up to the structure group, so shared maps must be intertwiners, and lattice gauge equivariant networks (Favoni et al., 2022; Kanwar et al., 2020; Boyda et al., 2021) transport features with physical link variables on a spacetime lattice. Here frames and generators are learned per token, and the lattice is the agents' communication graph.

Links, quivers, and sheaves. Group-valued edge variables with the endpoint law are the link fields of lattice gauge theory (Wilson, 1974; Kogut and Susskind, 1975; Kogut, 1979; Creutz, 1983), matrix links with a compact gauge group appear in linear lattice gauge theory (Wetterich, 2014), and linear maps on graph arrows modulo vertex groups are quiver representations (King, 1994; Douglas and Moore, 1996). Linear maps from vertex spaces into edge spaces are cellular sheaves (Hansen and Ghrist, 2019), used in discourse sheaves for agents with private opinions (Hansen and Ghrist, 2021) and in sheaf networks (Hansen and Gebhart, 2020; Bodnar et al., 2022; Barbero et al., 2022), and orthogonal edge transports underlie vector diffusion maps (Singer and Wu, 2012). The comparison and content maps here are sheaf-type data, the comparison cospan following the discourse-sheaf pattern, and the exponential link is the invertible, lattice-gauge member of the family, with a noncompact structure group.

Matrix-valued tokens. Token matrices appear in compositional matrix-space models (Rudolph and Giesbrecht, 2010; Baroni and Zamparelli, 2010; Mai et al., 2019), the matrix-vector recursive network (Socher et al., 2012), and multiplicative interactions (Jayakumar et al., 2020). There the matrix is an operator composed along the sequence; the frame U_i is a coordinate system for the token's own belief, whose gauge freedom leaves every prediction unchanged.

2 Statistical Agents on a Bundle over a Point

This section defines the agents, the four maps through which they communicate, and the attention operator and language model; Sections 3 to 6 establish their properties.

2.1 Principal frames and statistical associated bundles

Let $\mathcal{C}$ be a smooth contextual base, $G = \mathrm{GL}(K, \mathbb{R})$, and

$$\pi : \mathcal{P} \longrightarrow \mathcal{C} \tag{3}$$

a principal bundle of reference frames, on which G acts freely on the right, preserving π and transitively on each fiber, so one frame identifies a fiber with G. States attach through associated bundles (Nakahara, 2003; Frankel, 2011). With $\mathsf{V} = \mathbb{R}^K$ the defining representation, let

$$\mathcal{B} = \{\mathcal{N}(\mu, \Sigma) : \mu \in \mathsf{V},\ \Sigma \succ 0\}, \qquad \rho(g)\mathcal{N}(\mu, \Sigma) = \mathcal{N}(g\mu, g\Sigma g^\top). \tag{4}$$

The vector and Gaussian statistical bundles

$$E = \mathcal{P} \times_G \mathsf{V}, \qquad \mathcal{G}(E) = \mathcal{P} \times_\rho \mathcal{B}, \tag{5}$$

with the quotient convention

$$(n \cdot g, b) \sim (n, \rho(g)b) \tag{6}$$

and $v \mapsto gv$ in place of $\rho(g)$ for the vector bundle, carry latent values and distributions over those values, while the principal bundle carries gauge freedom.

The two fibers differ in kind. The fiber E_c of E is the sample space, a K-dimensional real vector space of latent values. A belief is instead a point of the statistical fiber $\mathcal{G}(E_c)$, in one frame the manifold $\mathbb{R}^K \times \mathrm{SPD}(K)$ of dimension $K + K(K+1)/2$, not a vector space because the open convex cone $\mathrm{SPD}(K)$ is not closed under subtraction. Its geometry is the Fisher metric (Amari and Nagaoka, 2000; Skovgaard, 1984), which at (μ, Σ) evaluates tangent vectors $(v, A), (w, B) \in \mathbb{R}^K \times \mathrm{Sym}(K)$ as

$$g_{(\mu,\Sigma)}\big((v, A), (w, B)\big) = v^\top \Sigma^{-1} w + \tfrac{1}{2} \operatorname{Tr}\big(\Sigma^{-1} A \Sigma^{-1} B\big). \tag{7}$$

The fiber is curved; for $K = 1$ it is the hyperbolic plane of curvature $-1/2$. The group acts by $(\mu, \Sigma) \mapsto (g\mu, g\Sigma g^\top)$, linearly on samples and isometrically on laws, since $(v, A) \mapsto (gv, gAg^\top)$ leaves both terms of Eq. (7) unchanged; linear maps between sample fibers act on beliefs by pushforward, invertible ones as Fisher isometries. The attention operator updates the mean linearly in the sample fiber and the covariance by congruence; it is not a geodesic or natural-gradient step on the statistical fiber.

An agent over a contextual region $\mathcal{U}_i$ selects a local principal section u_i and maintains belief and prior sections,

$$\begin{aligned} u_i &: \mathcal{U}_i \longrightarrow \mathcal{P}, \qquad \pi \circ u_i = \mathrm{id}_{\mathcal{U}_i}, \\ q_i, p_i &\in \Gamma\big(\mathcal{G}(E)|_{\mathcal{U}_i}\big), \end{aligned} \tag{8}$$

where Γ denotes smooth sections and (μ_i, Σ_i), $(\mu_{p,i}, \Sigma_{p,i})$ are coordinates in u_i. Given a reference trivialization σ_0 on the working domain, the frame U_i, defined by $u_i = \sigma_0 \cdot U_i^{-1}$, is the linear isomorphism $U_i(c) : \mathsf{V} \to E_c$ from σ_0- to u_i-coordinates, so $\xi_i = U_i^{-1}\mu_i$ are the σ_0-coordinates of the mean. On overlaps, agents may hold different beliefs at the same context, not local representatives of one common belief (Figure 1).

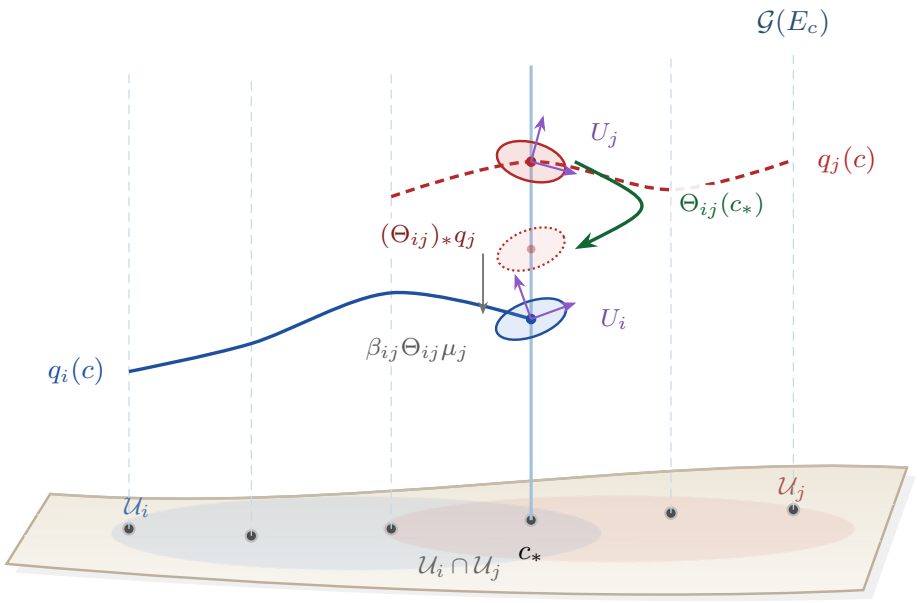


Figure 1: **Statistical sections, frames, and a group link.** Over a general contextual base $\mathcal{C}$, the curves are sections q_i, q_j of the statistical bundle over the overlapping domains of two agents. The token realization keeps only the fiber over c_* (solid line), where the glyphs carry the frames U_i, U_j; its communication graph is supplied separately. The green arrow applies the group link $\Theta_{ij} = U_i e^{L_{ij}} U_j^{-1}$, the dotted contour is the nondegenerate pushforward $(\Theta_{ij})_* q_j$, which the receiver's belief need not equal, and the gray arrow is the contribution $\beta_{ij}\Theta_{ij}\mu_j$ to the receiver's mean (exponential chart, one routing weight per edge; Theorem 7). Selective channels are drawn in Figure 2.

2.2 Agents at a single context

The token realization evaluates the construction over the zero-dimensional base $\mathcal{C} = \{c_*\}$, and each token position $i \in \mathcal{I} = \{0, \dots, N-1\}$ is an agent, as in targeted multi-agent communication through attention (Das et al., 2019). The per-token fibers E_i are a pullback along a constant map, so no connection, curvature, or topology of the base enters (Appendix A). Agents communicate along a directed graph of allowed sender-receiver pairs $j \to i$, causally $j \le i$; this communication graph is incidence data above c_* and carries the links of Section 3.

An agent is a triple (q_i, p_i, U_i) with frame $U_i : \mathsf{V} \to E_i$. Its belief and prior are points of the statistical fiber $\mathcal{G}(E_i)$ of the bundle of Eq. (5), in frame coordinates

$$q_i = \mathcal{N}(\mu_i, \Sigma_i), \qquad \Sigma_i \succ 0, \tag{9}$$

$$p_i = \mathcal{N}(\mu_{p,i}, \Sigma_{p,i}), \qquad \Sigma_{p,i} \succ 0, \tag{10}$$

with reference-coordinate representation

$$\widetilde{q}_i = (U_i^{-1})_* q_i = \mathcal{N}(\xi_i, C_i), \qquad \xi_i = U_i^{-1}\mu_i, \quad C_i = U_i^{-1}\Sigma_i U_i^{-\top}, \tag{11}$$

and an unrestricted positive-definite covariance.

A rechoice $u_i \mapsto u_i \cdot g_i^{-1}$ of the section of agent i, with σ_0 held fixed, changes the coordinates by Eq. (6):

$$(\mu_i, \Sigma_i, U_i) \mapsto (g_i\mu_i, g_i\Sigma_i g_i^{\top}, g_i U_i), \qquad g_i \in \mathrm{GL}(K). \tag{12}$$

These independent passive changes leave the reference moments (ξ_i, C_i) unchanged and hold the reference-space interface parameters fixed; covectors transform contragrediently. A change of σ_0 is instead the common reference change $U_i \mapsto U_i a$, which also transforms the shared interface maps (Appendix A).

Each token type $c \in \{1, \dots, V\}$ carries a learned prior and frame coefficients,

$$p_c = \mathcal{N}(\mu_p^c, \Sigma_p^c), \qquad \phi_c \in \mathfrak{gl}(K), \tag{13}$$

and each agent starts from the prior of its token, $q_i^{(0)} = p_{x_i}$. This prior bank is a Gaussian embedding table (Vilnis and McCallum, 2015) extended by frame coefficients.

Each frame composes the token coefficients with a learned positional generator $Y_i \in \mathfrak{gl}(K, \mathbb{R})$, $U_i = \exp(\phi_{x_i})\exp(Y_i)$, of positive determinant. In the evaluated runs both are block diagonal with B blocks of widths d_b, so the frames lie in $\prod_b \mathrm{GL}^+(d_b)$ (Supplement S7); Appendix A records the restrictions of this chart.

2.3 Communication: whom to hear and what to say

A receiver must decide whom to hear, and each sender's contribution must be put into a form the receiver can absorb; one linear map cannot do both. A head therefore carries four maps on the reference space, split by side and by role:

	Relevance: whom to hear	Content: what is said
Sender j	key map $S_{K,h}$, emits $S_{K,h}\xi_j$	sending map T_h, emits $T_h\xi_j$
Receiver i	query map $S_{Q,h}$, forms $S_{Q,h}\xi_i$	receiving map $\widetilde{O}_h$, writes into ξ_i

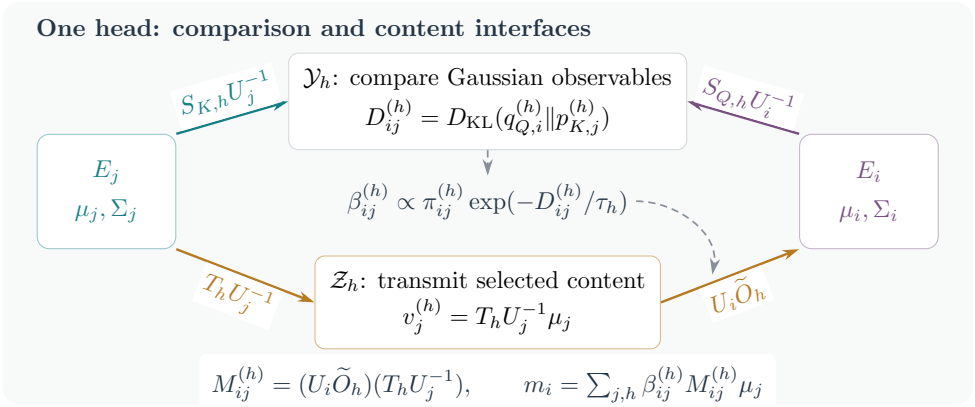


Figure 2: **Comparison and content interfaces of one head.** Key and sending maps act on the sender's side, query and receiving maps on the receiver's side. Query and key send Gaussian observables into $\mathcal{Y}_h$, where their divergence sets the routing weight (dashed arrows: scalar control, not a state map; Section 6.1); the content maps carry the mean through $\mathcal{Z}_h$ and compose to the head map $M_{ij}^{(h)}$, singular when $s_h < K$.

The sender emits two partial representations of its belief: its key $(S_{K,h})_* \widetilde{q}_j$, what makes it findable, and its content $T_h \xi_j$, what it says. The receiver forms a query $(S_{Q,h})_* \widetilde{q}_i$, what it is looking for, which is never transmitted, and writes received content into its state with $\widetilde{O}_h$. The routing weight $\beta_{ij}^{(h)}$ is the decision of whom to hear; query and key are what it is computed from, and nothing they measure enters the receiver's state. The interface maps are a protocol shared by all agents of a layer; the private frames translate to the reference space, the sender reading through U_j^{-1} and the receiver writing through U_i. None of the four maps selects a subset of coordinates or of the frame: each reads linear combinations of all K reference coordinates, T_h discards $\ker T_h$, and $\widetilde{O}_h$ writes into its image without reconstructing the sender's state. The maps correspond to the transformer projections W_Q, W_K, W_V, W_O (Vaswani et al., 2017), with $\widetilde{O}_h = O_h$ in the affine chart, and separating an addressing signature from a transmitted value is the message structure of Das et al. (2019).

A layer ℓ maps beliefs $q_i^{(\ell)} = \mathcal{N}(\mu_i^{(\ell)}, \Sigma_i^{(\ell)})$, with reference moments $\xi_i^{(\ell)}, C_i^{(\ell)}$, to $q_i^{(\ell+1)}$ through H heads (Figure 2), which need not partition the state or coincide with frame blocks. Head h has a comparison space, where compatibility is measured, and a content space, through which the message passes,

$$\mathcal{Y}_h = \mathbb{R}^{r_h}, \qquad \mathcal{Z}_h = \mathbb{R}^{s_h}, \tag{14}$$

both shared by all agents, and interfaces of the declared types

$$S_{Q,h}^{(\ell)}, S_{K,h}^{(\ell)} : \mathsf{V} \longrightarrow \mathcal{Y}_h, \tag{15}$$

$$T_h^{(\ell)} : \mathsf{V} \longrightarrow \mathcal{Z}_h, \qquad O_h^{(\ell)} : \mathcal{Z}_h \longrightarrow \mathsf{V}, \tag{16}$$

with $S_{Q,h}$ and $S_{K,h}$ of full row rank, so $r_h \le K$. The applied receiving map $\widetilde{O}_h^{(\ell)} : \mathcal{Z}_h \to \mathsf{V}$ is $O_h^{(\ell)}$ in the affine chart and is evaluated from the stored (T, O) of all heads in the other charts of Eq. (43). Layer indices are suppressed for a fixed layer, matrices act on column vectors, and ij maps sender j to receiver i. Query and key laws share the comparison space:

$$q_{Q,i}^{(\ell,h)} = \mathcal{N}\big(S_{Q,h}^{(\ell)}\xi_i^{(\ell)}, S_{Q,h}^{(\ell)}C_i^{(\ell)}S_{Q,h}^{(\ell)\top}\big), \tag{17}$$

$$p_{K,j}^{(\ell,h)} = \mathcal{N}\big(S_{K,h}^{(\ell)}\xi_j^{(\ell)}, S_{K,h}^{(\ell)}C_j^{(\ell)}S_{K,h}^{(\ell)\top}\big), \tag{18}$$

$$D_{ij}^{(\ell,h)} = D_{\mathrm{KL}}\big(q_{Q,i}^{(\ell,h)} \| p_{K,j}^{(\ell,h)}\big). \tag{19}$$

Since $C_i^{(\ell)} \succ 0$ and each comparison map S has full row rank, $y^\top SCS^\top y = (S^\top y)^\top C(S^\top y) > 0$ for $y \neq 0$: the comparison uses full nondegenerate reduced covariances, not a singular K-dimensional projection. For $S_{Q,h} = S_{K,h}$ every receiver attains the minimum $D_{ii}^{(h)} = 0$ at itself; otherwise identical beliefs can have positive cost, as the generally nonsymmetric form $W_Q^\top W_K$ allows in dot-product attention. Even when square, these maps are learned parameters shared by all agents, not gauge transformations.

With the local measurements $A_{Q,i,h} = S_{Q,h}U_i^{-1}$ and $A_{K,j,h} = S_{K,h}U_j^{-1}$, two agents are compared through the cospan

$$E_i \xrightarrow{A_{Q,i,h}} \mathcal{Y}_h \xleftarrow{A_{K,j,h}} E_j. \tag{20}$$

On one edge this is a cellular sheaf with vertex stalks E_i, E_j, edge stalk $\mathcal{Y}_h$, and restriction maps $A_{Q,i,h}, A_{K,j,h}$ (Hansen and Ghrist, 2019), close to the discourse sheaves of opinion dynamics (Hansen and Ghrist, 2021), except that an agent's restriction map depends on its role on the edge and is one reference map dressed by its frame. A cospan chooses no map back from $\mathcal{Y}_h$ to E_i; the content maps instead compose to a map from sender to receiver fiber,

$$E_j \xrightarrow{T_h U_j^{-1}} \mathcal{Z}_h \xrightarrow{U_i \widetilde{O}_h} E_i,$$

the head map $M_{ij}^{(h)}$ of Section 2.4. The following proposition, proved in Appendix A, states when a comparison can itself be written through a map from sender to receiver.

Proposition 1 (Comparison through a link) *Fix a head, suppress h, and let q_i and q_j be nondegenerate Gaussian beliefs with reference laws $\widetilde{q}_i = (U_i^{-1})_* q_i$ and $\widetilde{q}_j = (U_j^{-1})_* q_j$.*

1. *If $r_h = K$ and S_Q and S_K are invertible, then*

$$D_{ij} = D_{\mathrm{KL}}\big((S_Q)_*\widetilde{q}_i \| (S_K)_*\widetilde{q}_j\big) = D_{\mathrm{KL}}\big(q_i \| (\Theta_{ij}^{\mathrm{cmp}})_* q_j\big), \qquad \Theta_{ij}^{\mathrm{cmp}} = U_i S_Q^{-1} S_K U_j^{-1}. \tag{21}$$

The comparison link $\Theta_{ij}^{\mathrm{cmp}}$ is group valued, with reference representative $\mathring{\Theta}^{\mathrm{cmp}} = S_Q^{-1}S_K \in \mathrm{GL}(K)$ common to all edges. It obeys the endpoint law $\Theta_{ij}^{\mathrm{cmp}\prime} = g_i \Theta_{ij}^{\mathrm{cmp}} g_j^{-1}$ under Eq. (12); it is unchanged by the redundancy $(S_Q, S_K) \mapsto (AS_Q, AS_K)$ with $A \in \mathrm{GL}(K)$ and by the common reference change, under which its reference representative transforms by the adjoint action. It is an exponential group link $U_i e^{L^{\mathrm{cmp}}} U_j^{-1}$, with one comparison generator $L^{\mathrm{cmp}} \in \mathfrak{gl}(K, \mathbb{R})$ for all edges, exactly when $S_Q^{-1}S_K$ has a real logarithm L^{cmp}. For $S_Q = S_K$ it is the flat identification $\Omega_{ij} = U_i U_j^{-1}$, and $D_{ij} = D_{\mathrm{KL}}(q_i \| (\Omega_{ij})_* q_j)$.

2. *If* $M : E_j \to E_i$ *is linear with* $\operatorname{rank} M < K$*, then* $M_* q_j = \mathcal{N}(M\mu_j, M\Sigma_j M^\top)$ *is degenerate and*

$$D_{\mathrm{KL}}\big(q_i \| M_* q_j\big) = D_{\mathrm{KL}}\big(M_* q_j \| q_i\big) = +\infty. \tag{22}$$

In particular, this holds for every head map $M_{ij}^{(h)}$ *with* $s_h < K$*.*

Two square invertible comparison maps thus define a group link, and the routing cost is the divergence of the receiver's belief from the sender's belief transported by it, the flat identification being the case $S_Q = S_K$; Section 6.1 states when it is the content link of Section 3. Item 2 gives the role of $S_{Q,h}$: a partial channel can be compared with a belief only after the receiver is measured in the channel's own space, where both laws are nondegenerate and D_{ij} is finite for every $r_h \le K$. Comparing a whole belief against a partial message would penalize disagreement on everything the sender did not address. For $r_h < K$ the cospan induces only a state-dependent comparison map (Supplement S2.3), and the evaluated predictor uses only D_{ij}. When the key map factors through the sending map, $S_{K,h} = A_h T_h$, the receiver selects sources by the content itself; every reported run uses independent key and sending maps.

2.4 The attention operator and the language model

One layer combines invariant routing weights with covariant content maps: each receiver adds a weighted message from its allowed senders to its mean.

Routing. At fixed input beliefs and interface maps, source selection minimizes the head-wise objective

$$\mathcal{A}_i^{(\ell,h)}(\beta) = \sum_j \beta_j D_{ij}^{(\ell,h)} + \tau_h \sum_j \beta_j \log \frac{\beta_j}{\pi_{ij}^{(h)}}, \qquad \sum_j \beta_j = 1, \tag{23}$$

on the support of $\pi_i^{(h)}$, causally $\mathcal{N}_i = \{0, \dots, i\}$ for every head. A normalization multiplier gives

$$\beta_{ij}^{(\ell,h)} = \operatorname{softmax}_j \left(-D_{ij}^{(\ell,h)}/\tau_h + \log \pi_{ij}^{(h)} \right), \tag{24}$$

the unique categorical minimizer for finite costs, nonempty support, and positive temperature. The source prior sets the communication graph and any positional preference; the temperature is an independent positive scalar (Section 5). Section 6 gives the probabilistic status of the optimizer, and Appendix D the causal, window, ALiBi, and relative-position priors.

Content. The sending map $T_h^{(\ell)} \in \mathbb{R}^{s_h \times K}$ extracts content, and the receiving map $\widetilde{O}_h^{(\ell)} \in \mathbb{R}^{K \times s_h}$ places the head's aggregate in the full state space:

$$v_j^{(\ell,h)} = T_h^{(\ell)} \xi_j^{(\ell)}, \tag{25}$$

$$z_i^{(\ell,h)} = \sum_j \beta_{ij}^{(\ell,h)} v_j^{(\ell,h)}, \tag{26}$$

$$m_i^{(\ell)} = U_i \sum_{h=1}^{H} \widetilde{O}_h^{(\ell)} z_i^{(\ell,h)}, \tag{27}$$

$$\mu_i^{\mathrm{att},(\ell)} = c_{\mathrm{res}} \mu_i^{(\ell)} + m_i^{(\ell)}, \qquad c_{\mathrm{res}} \in \{0, 1\}. \tag{28}$$

For $c_{\mathrm{res}} = 1$ the summed message is added to the mean, as in a transformer residual; for $c_{\mathrm{res}} = 0$ it replaces the mean. The coefficient is architectural, not a belief-gradient step size; its relation to the link charts is discussed with Theorem 7.

For a fixed layer, the content maps of head h composed with the frames give the head map

$$M_{ij}^{(h)} = (U_i \widetilde{O}_h)(T_h U_j^{-1}) = U_i M_h U_j^{-1} : E_j \longrightarrow E_i, \qquad M_h = \widetilde{O}_h T_h, \tag{29}$$

whose factors need not be square; in the affine chart it is $U_i O_h T_h U_j^{-1}$, derived in Section 3 as the first-order channel of a group link. Under Eq. (12), direct substitution gives

$$(T_h U_j^{-1})' = T_h U_j^{-1} g_j^{-1}, \qquad (U_i \widetilde{O}_h)' = g_i U_i \widetilde{O}_h, \qquad M_{ij}^{(h)\prime} = g_i M_{ij}^{(h)} g_j^{-1}, \tag{30}$$

an endpoint law carried by the frames, with the reference transfer M_h unchanged. For invariant scalar routing weights, the routed edge map and the message are

$$\mathcal{M}_{ij} = \sum_h \beta_{ij}^{(h)} M_{ij}^{(h)} = U_i \left(\sum_h \beta_{ij}^{(h)} M_h \right) U_j^{-1} = U_i \widetilde{O} \boldsymbol{\beta}_{ij} T U_j^{-1}, \tag{31}$$

$$m_i = \sum_j \mathcal{M}_{ij} \mu_j, \tag{32}$$

where $T = [T_1; \cdots; T_H]$ and $\widetilde{O} = [\widetilde{O}_1 \ \cdots \ \widetilde{O}_H]$ stack the heads, $\boldsymbol{\beta}_{ij} = \bigoplus_h \beta_{ij}^{(h)} I_{s_h}$, and $\mathring{\mathcal{M}}_{ij} = \widetilde{O} \boldsymbol{\beta}_{ij} T = \sum_h \beta_{ij}^{(h)} M_h$ is the reference representative of the routed edge map. Since $T_h \xi_j = T_h U_j^{-1} \mu_j$, this is the message of Eq. (28). With invariant routing, Eq. (30) gives $\mathcal{M}'_{ij} = g_i \mathcal{M}_{ij} g_j^{-1}$ and $m'_i = g_i m_i$, so the attention update $c_{\mathrm{res}} \mu_i + m_i$ is covariant (Theorem 11).

Depth. A model stacks N_ℓ layers with maps $S_Q^{(\ell)}, S_K^{(\ell)}, T^{(\ell)}, O^{(\ell)}$; the frame field may be shared across depths. An affine normalization and a coordinate residual MLP after attention give

$$\bar{\mu}_i^{(\ell)} = \mathrm{LayerNorm}_{\gamma,b}\big(\mu_i^{\mathrm{att},(\ell)}\big), \tag{33}$$

$$\mu_i^{(\ell+1)} = \bar{\mu}_i^{(\ell)} + W_2^{(\ell)} \, \mathrm{GELU}\big(W_1^{(\ell)} \bar{\mu}_i^{(\ell)} + b_1^{(\ell)}\big) + b_2^{(\ell)}, \tag{34}$$

with layer-normalization parameters γ, b (Ba et al., 2016) shared across depths, per-depth MLP parameters, and the Gaussian error linear unit (Hendrycks and Gimpel, 2016). Under

covariance passthrough, $\Sigma_i^{(\ell+1)} = \Sigma_i^{(\ell)}$, the covariance is a routing state rather than propagated uncertainty of the nonlinear mean map; under moment matching, attention returns the covariance of the predictive kernel of Theorem 15, passed unchanged through normalization and MLP. Coordinatewise activations and a fixed affine normalization do not in general commute with $\mathrm{GL}(K)$, so the covariance of Theorem 11 does not extend automatically to this block; Corollary 12 gives its normal form, and Supplement S8.2.2 the Gaussian-closure restriction.

Prediction. For input tokens $x_0, \dots, x_{N-1}$, the sequence model has the normalized factorization

$$p_\theta(x_0, \dots, x_N) = p_\theta(x_0) \prod_{i=0}^{N-1} p_\theta(x_{i+1} \mid x_{\le i}), \tag{35}$$

completed by a proper initial-token law, and a linear categorical decoder specifies the next-token factors from the final mean,

$$p_\theta(x_{i+1} = v \mid x_{\le i}) = \mathrm{softmax}_v \left(W_{\mathrm{out}} \mu_i^{(N_\ell)} + b_{\mathrm{out}} + \kappa_{\mathrm{uni}} \log p_{\mathrm{uni}} \right), \tag{36}$$

with a positive vocabulary prior p_{uni} of weight κ_{uni}. This decoder is a Gibbs readout of shared-covariance Gaussian templates (Appendix E); the token-bank decoder of Supplement S8.3 is another choice. Next-token cross-entropy trains the composed predictor through all differentiable components; the observed next token is a target, not an input to the causal state (Algorithm 1).

3 Group-Valued Links and Multi-Head Attention

The content map $M_{ij}^{(h)}$ of Eq. (29) is a composite of site frames and shared maps, singular when $s_h < K$, whereas a lattice link variable is a group element whose logarithmic coordinate is the algebra-valued potential. This section places such links on the communication graph and derives from them a head as the channel term of a group link (Section 3.2), the routed message as a routed oblique projection followed by link transport (Section 3.3), and the standard transformer layer as the affine chart with a full residual (Section 3.4).

3.1 Group links on the communication graph

Definition 2 (Exponential group link) *For a generator $L_{ij} \in \mathfrak{gl}(K, \mathbb{R})$ expressed in reference coordinates, the group link with Lie-algebra generator L_{ij}, or exponential group link, from sender j to receiver i is*

$$\Theta_{ij} = U_i \exp(L_{ij}) U_j^{-1} : E_j \longrightarrow E_i, \tag{37}$$

and its reference representative is $\mathring{\Theta}_{ij} = U_i^{-1} \Theta_{ij} U_j = \exp(L_{ij})$.

The evaluated model shares one generator per layer, $L^{(\ell)} = O^{(\ell)} T^{(\ell)}$, written L when the layer is fixed; edge-dependent generators are the latent-link extension of Supplement S6.

Proposition 3 (Endpoint law and invertibility) *Let $L_{ij} \in \mathfrak{gl}(K, \mathbb{R})$.*

1. *Under the passive site action of Eq.* (12)*, with the reference generator held fixed,* $\Theta'_{ij} = g_i \Theta_{ij} g_j^{-1}$.

2. *Under the common reference change* $U_i \mapsto U_i a$*, the same local link is represented by the generator* $a^{-1} L_{ij} a$*, the adjoint action of* $\mathrm{GL}(K)$ *on its Lie algebra.*

3. $\det \Theta_{ij} = \det(U_i U_j^{-1}) e^{\mathrm{tr}\, L_{ij}} \neq 0$. *Every exponential group link is invertible, including links with singular generators, and* $\Theta_{ij}^{-1} = U_j e^{-L_{ij}} U_i^{-1}$.

4. *A matrix* $A \in \mathrm{GL}(K, \mathbb{R})$ *equals* e^L *for some real* L *exactly when, for each negative eigenvalue of* A *and each block size, the Jordan blocks of that eigenvalue and size occur an even number of times. Every such* A *lies in the identity component* $\mathrm{GL}^+(K)$*, and for* $K \geq 2$ *not every element of* $\mathrm{GL}^+(K)$ *is of this form.*

Item 4 is the real-logarithm criterion of Culver (1966), and item 1 is the endpoint law of a lattice link variable (Wilson, 1974; Kogut, 1979; Creutz, 1983); the proof is in Appendix B.1.

The flat identification

$$\Omega_{ij} = U_i U_j^{-1}, \qquad \mu_j \mapsto \Omega_{ij} \mu_j, \tag{38}$$

is the exponential group link with zero generator and the head map of a head with $M_h = I_K$.

Remark 4 (Flat reference identification) *For any invertible frames and any finite sequence of sites,* $\Omega_{ij}\Omega_{jk} = U_i U_k^{-1} = \Omega_{ik}$*, so an ordered product of* Ω *maps depends only on its endpoints and every closed product equals* I_K*. The reference identification is pure gauge: it is the coboundary of the frame field.*

Matrix-valued and group-valued links. The edge objects of Table 1 all obey the endpoint law of Eq. (30), a property of $\mathrm{Hom}(E_j, E_i)$, so covariance does not distinguish matrix maps from group links, and for $G = \mathrm{GL}(K)$ the two coincide exactly on the open dense set of invertible matrices. Group links form a groupoid: loop products are invertible with conjugacy-class invariants, the inverse edges Θ_{ij}^{-1} cost no parameters and give holonomy to the causal graph (Section 3.5), and the pushforward of a nondegenerate Gaussian is a Fisher isometry. Matrix maps form a monoid: rank cannot increase along a path, reverse edges must be declared, and a singular map leaves the statistical manifold (Proposition 1). The exponential reaches only $\mathrm{GL}^+(K)$ (Proposition 3), and its generator is a linear coordinate for priors, natural gradients, and latent inference (Supplement S6) along which no step crosses the singular set, as an additive step can (Supplement S1.1); this concerns the optimization geometry, not the conditioning of trained links (Section 7.4). Since $M_{ij}^{(h)} N$ is singular for every N when $s_h < K$, no group element factors through a selective channel; the two relate through the Lie algebra (Section 3.2), and a routed sum of heads can be invertible (Proposition S5, Theorem 7). With frames in $G = \prod_b \mathrm{GL}(d_b)$, as evaluated, the links take values in $\mathrm{GL}^+(K)$, not in G, because the full-width interface maps couple the blocks.

Lattice and sheaf data. The lattice is the communication graph above c_*, with agents as vertices and allowed source relations as edges; it discretizes no base manifold and is not the single-site reduction of large-N gauge theory, whose links are labeled by collapsed lattice directions (Eguchi and Kawai, 1982). Invertible links with the endpoint law are

Edge object	Symbol	Values and invertibility	Implemented as
Generator	L_{ij}	$\mathfrak{gl}(K,\mathbb{R})$ in reference coordinates; unchanged by the passive action, adjoint under a reference change	one $L^{(\ell)} = O^{(\ell)}T^{(\ell)}$ per layer
Group link	$\Theta_{ij} = U_i e^{L_{ij}} U_j^{-1}$, $\mathring{\Theta}_{ij} = e^{L_{ij}}$	invertible; $\mathring{\Theta}_{ij} \in \mathrm{GL}^+(K)$	$U_i e^{L} U_j^{-1}$, exponential chart only
Flat identification	$\Omega_{ij} = U_i U_j^{-1}$	invertible; $\Omega_{ij}\Omega_{jk} = \Omega_{ik}$, pure gauge	the frame field
Comparison link	$\Theta^{\mathrm{cmp}}_{ij} = U_i S_Q^{-1} S_K U_j^{-1}$	invertible; requires $r_h = K$ (Proposition 1)	not formed ($r_h < K$)
Head map	$M^{(h)}_{ij} = U_i M_h U_j^{-1}$	rank at most s_h; singular if $s_h < K$	every chart
Routed edge map	$\mathcal{M}_{ij} = U_i \widetilde{O} \boldsymbol{\beta}_{ij} T U_j^{-1}$, $\mathring{\mathcal{M}}_{ij} = \widetilde{O}\boldsymbol{\beta}_{ij}T$	rank at most $\min(K, \bar{s})$; in $U_i\mathrm{GL}^+(K)U_j^{-1}$ for tiled heads in the exponential chart with all $\beta^{(h)}_{ij} > 0$	the evaluated operator

Table 1: Edge objects of the construction, with $\bar{s} = \sum_h s_h$ and layer indices suppressed. All of them obey the endpoint law $\mathcal{K}_{ij} \mapsto g_i \mathcal{K}_{ij} g_j^{-1}$ under Eq. (12); they differ in composition, invertibility, and the component of $\mathrm{GL}(K)$ they reach.

lattice-gauge data (Wilson, 1974; Kogut, 1979; Creutz, 1983) and, as edge variables under independent vertex groups, bifundamental fields of a quiver gauge theory (Douglas and Moore, 1996). The singular head maps and comparison maps are quiver-representation and cellular-sheaf data (King, 1994; Hansen and Ghrist, 2019), with the comparison cospan of Eq. (20) as the sheaf form; group links are their invertible restriction.

3.2 Multi-head attention as the channel term of a group link

A generator built from the stored value and output maps has an exponential in closed form through the entire function

$$\varphi_1(Z) = \sum_{n\geq 1} \frac{Z^{n-1}}{n!}, \qquad \varphi_1(Z) = (e^Z - I)Z^{-1} \text{ for invertible } Z, \tag{39}$$

the first of the φ-functions of exponential integrators (Hochbruck and Ostermann, 2010).

Lemma 5 (Exponential of a low-rank generator) *For $A \in \mathbb{R}^{K\times s}$ and $B \in \mathbb{R}^{s\times K}$,*

$$\exp(AB) = I_K + A\varphi_1(BA)B. \tag{40}$$

Proof For $n \geq 1$, regrouping the product gives $(AB)^n = A(BA)^{n-1}B$. Summing the exponential series term by term gives $I_K + A\big(\sum_{n\geq 1}(BA)^{n-1}/n!\big)B$, and the inner series converges for every $s \times s$ matrix. ∎

Proposition 6 (Flat transport plus a channel term) *For one head with stored maps T_h, O_h and the generator $L_{ij} = O_h T_h$,*

$$\Theta_{ij} = \Omega_{ij} + U_i O_h \varphi_1(T_h O_h) T_h U_j^{-1}. \tag{41}$$

1. $\exp(O_h T_h)$ *acts as the identity on* $\ker T_h$*, so on the subspace* $U_j \ker T_h$ *that the head does not read,* Θ_{ij} *coincides with the flat identification* Ω_{ij}*.*
2. *The image of* $\exp(O_h T_h) - I$ *lies in* $\operatorname{im} O_h$*, so* Θ_{ij} *differs from* Ω_{ij} *only in directions that the head writes.*
3. *For* $L_{ij} = s O_h T_h$*,* $\Theta_{ij} = \Omega_{ij} + s U_i O_h T_h U_j^{-1} + \mathcal{O}(s^2)$*, so the affine-chart head map* $U_i O_h T_h U_j^{-1}$ *is the first-order channel.*
4. $\det \Theta_{ij} = \det(U_i U_j^{-1}) e^{\operatorname{tr}(T_h O_h)}$*.*

The same statements hold for the stacked generator $L = OT = \sum_h O_h T_h$*, with* $O = [O_1 \ \cdots \ O_H] \in \mathbb{R}^{K \times \bar{s}}$*,* $T = [T_1; \cdots; T_H] \in \mathbb{R}^{\bar{s} \times K}$*, and* $\bar{s} = \sum_h s_h$*.*

Proof Equation (41) is Lemma 5 with $A = O_h$ and $B = T_h$, conjugated by the frames. Items 1 and 2 are read off from the correction term, which has the form $O_h(\cdots)T_h$. Item 3 follows from $\varphi_1(sZ) = I + \mathcal{O}(s)$, and item 4 from $\det e^L = e^{\operatorname{tr} L}$ with $\operatorname{tr}(O_h T_h) = \operatorname{tr}(T_h O_h)$. ■

The link is the identity on the unread complement plus a channel term equal to the affine-chart head map at first order: multi-head attention is the channel term of a group link whose zeroth order is the flat identification. Along one routing row the two orders are the messages of complete-state alignment and of affine-chart prediction (Appendix B.2).

3.3 Routed transport

The group law does not choose an aggregation over sources, and placing the routing inside the exponent, $L_{ij} = O \boldsymbol{\beta}_{ij} T$, leaves it unspecified (Remark S4). The construction keeps categorical routing outside the exponent with the routing-independent stacked generator $L = OT$, so that $\beta_{ij}^{(h)}$ remains the probability that head h selects source j (Theorem 15). The heads tile the reference space, $\bar{s} = K$, and the stacked sending map $T = [T_1; \cdots; T_H] \in \mathbb{R}^{K \times K}$ is invertible. Write $\boldsymbol{\beta}_{ij} = \bigoplus_h \beta_{ij}^{(h)} I_{s_h}$ for the head-diagonal routing matrix, let Δ_h be the coordinate projector onto the head-h block of $\mathbb{R}^K$, and put

$$\Pi_h = T^{-1} \Delta_h T, \qquad \Pi(\boldsymbol{\beta}) = T^{-1} \boldsymbol{\beta} T. \tag{42}$$

The stored factors define the affine, first-order, and exponential charts of the implementation through their effective receiving maps,

$$\widetilde{O} = O, \qquad \widetilde{O} = T^{-1} + O, \qquad \widetilde{O} = T^{-1} + O\varphi_1(TO) = e^{OT} T^{-1}. \tag{43}$$

The affine chart keeps only the first-order channel, so $O = 0$ means zero content; the first-order and exponential charts are centered at the head-routed flat transport and differ from second order in O on; only the exponential chart keeps the effective transfer in $\mathrm{GL}^+(K)$. Later statements use the content maps only through $\widetilde{O}_h$ and T_h and so hold in every chart; initialization and φ_1 are treated in Appendix B.7.

Theorem 7 (Routed transport) *Assume tiled heads, $\bar{s} = K$, with an invertible stacked sending map T, routing outside the exponent, and the shared generator $L = OT$, and let Π_h and $\Pi(\boldsymbol{\beta})$ be as in Eq.* (42)*.*

1. *Oblique head projectors: $\Pi_h = (T^{-1})_{:,h} T_h$, $\Pi_h \Pi_g = \delta_{hg} \Pi_h$, $\sum_h \Pi_h = I_K$, $\ker \Pi_h = \ker T_h$, $\operatorname{im} \Pi_h = \bigcap_{g \neq h} \ker T_g$, and*

$$\Pi(\boldsymbol{\beta}_{ij}) = T^{-1} \boldsymbol{\beta}_{ij} T = \sum_h \beta_{ij}^{(h)} \Pi_h. \tag{44}$$

   The matrix $\Pi(\boldsymbol{\beta}_{ij})$ is diagonalizable with eigenvalue $\beta_{ij}^{(h)}$ on $\operatorname{im} \Pi_h$, which has dimension s_h, so $\det \Pi(\boldsymbol{\beta}_{ij}) = \prod_h (\beta_{ij}^{(h)})^{s_h}$.

2. *Charts: the effective receiving maps of Eq.* (43) *satisfy $\widetilde{O} T = f(L)$ with $f(L) = L$, $I_K + L$, and e^L in the affine, first-order, and exponential charts. In the exponential chart*

$$\widetilde{O} = e^L T^{-1} = T^{-1} + O \varphi_1(TO), \qquad T \widetilde{O} = e^{TO}, \tag{45}$$

   so $\det(T\widetilde{O}) = e^{\operatorname{tr} L} > 0$.

3. *Routed transport: in each chart the routed edge map of Eq.* (32) *and the message are*

$$\begin{aligned} \mathcal{M}_{ij} &= U_i f(L) \Pi(\boldsymbol{\beta}_{ij}) U_j^{-1}, \\ m_i &= \sum_j U_i f(L) \Pi(\boldsymbol{\beta}_{ij}) U_j^{-1} \mu_j = \sum_h \sum_j \beta_{ij}^{(h)} U_i \widetilde{O}_h T_h U_j^{-1} \mu_j. \end{aligned} \tag{46}$$

4. *Group-link factorization: in the exponential chart, with $\Theta_{ij} = U_i e^L U_j^{-1}$,*

$$\mathcal{M}_{ij} = \Theta_{ij} U_j \Pi(\boldsymbol{\beta}_{ij}) U_j^{-1}, \tag{47}$$

   and the reference representative $\mathring{\mathcal{M}}_{ij} = U_i^{-1} \mathcal{M}_{ij} U_j$, which equals $e^L \Pi(\boldsymbol{\beta}_{ij})$, has determinant $e^{\operatorname{tr} L} \prod_h (\beta_{ij}^{(h)})^{s_h}$. If $\beta_{ij}^{(h)} > 0$ for every head, $\mathcal{M}_{ij}$ is invertible and $\mathring{\mathcal{M}}_{ij} \in \mathrm{GL}^+(K)$. If every head uses the same routing, $\beta_{ij}^{(h)} = \beta_{ij}$, as always holds for $H = 1$, the message is

$$m_i = \sum_j \beta_{ij} \Theta_{ij} \mu_j \tag{48}$$

   in the exponential chart, $m_i = \sum_j \beta_{ij} U_i L U_j^{-1} \mu_j$ in the affine chart, and $m_i = \sum_j \beta_{ij} (\Omega_{ij} + U_i L U_j^{-1}) \mu_j$ in the first-order chart.

5. *Orders: replacing O by sO at fixed T replaces L by sL, and the exponential-chart receiving map becomes*

$$\widetilde{O}(s) = T^{-1} + sO \varphi_1(sTO) = T^{-1} + \sum_{n \geq 1} \frac{s^n}{n!} O (TO)^{n-1}, \tag{49}$$

which converges for every s. The order-n term of $\mathring{\mathcal{M}}_{ij}(s) = e^{sL}\Pi(\boldsymbol{\beta}_{ij})$ at $s = 1$ is $L^n\Pi(\boldsymbol{\beta}_{ij})/n!$. It equals $\sum_h \beta_{ij}^{(h)}\Pi_h$ at order zero, $O\boldsymbol{\beta}_{ij}T$ at order one, and at order two

$$\tfrac{1}{2}O(TO)\boldsymbol{\beta}_{ij}T = \tfrac{1}{2}\sum_{g,h} O_g(T_gO_h)\beta_{ij}^{(h)}T_h. \tag{50}$$

Under shared routing the order-zero message is $\sum_j \beta_{ij}\Omega_{ij}\mu_j$.

The proof is in Appendix B.3.

Item 3 is the central identity: each head selects source j with probability $\beta_{ij}^{(h)}$ and keeps its oblique component $\Pi_h\xi_j$, which no other head reads, and $f(L)$, dressed by the frames, transports the routed components into the receiver. In the exponential chart this transport is the group link (Eq. (47)), and softmax routing with finite costs makes every routed edge map invertible on the common support of the heads' priors. For shared routing, Eq. (48) is the gauge-covariant hopping term of lattice gauge theory (Kogut, 1979; Creutz, 1983) with row-stochastic, state-dependent amplitudes, so the correspondence holds at fixed routing. In the order expansion of item 5, order zero is a head-routed flat transport, $U_i\sum_h\sum_j \beta_{ij}^{(h)}\Pi_h\xi_j$, which includes the receiver through $\beta_{ii}^{(h)}$ and is not the residual $c_{\mathrm{res}}\mu_i$; order one, $U_i\sum_j O\boldsymbol{\beta}_{ij}T\xi_j$, is the standard multi-head value path (Section 5.3); order two, Eq. (50), is cross-head mixing, in which the block routed by head h is written by O_h, read by every head g through T_gO_h, and rewritten by O_g without a second routing, and order n makes $n-1$ such round trips with weight $1/n!$. The additive residual connection (He et al., 2016) has no counterpart in tiled aggregation (Remark 16).

3.4 Charts of one group and the standard transformer

The three charts are coordinates on one group, not successive approximations: on a specified logarithm domain $\Omega_{ij} + U_iO_{\mathrm{lin}}TU_j^{-1}$ is exactly an exponential group link (Proposition 17), and for tiled heads the three represent the same mean maps up to the real-logarithm domain (Appendix B.3).

The standard transformer layer. Take trivial frames and H heads with value maps $W_V^{(h)} \in \mathbb{R}^{d_h\times K}$ and output maps $W_O^{(h)} \in \mathbb{R}^{K\times d_h}$, $Hd_h = K$, stacked as $T = [W_V^{(1)};\cdots;W_V^{(H)}]$ and $O = [W_O^{(1)}\ \cdots\ W_O^{(H)}]$, so that $OT = \sum_h W_O^{(h)}W_V^{(h)}$; in the affine chart $\widetilde{O} = O$ is the output projection. At fixed routing weights the residual multi-head layer is linear in the joint state $\boldsymbol{\xi} = (\xi_0,\dots,\xi_{N-1}) \in \mathbb{R}^{NK}$,

$$\boldsymbol{\xi} \longmapsto (I_{NK} + \mathfrak{L}_\beta)\boldsymbol{\xi}, \qquad \mathfrak{L}_\beta = \sum_h B^{(h)}\otimes M_h \in \mathfrak{gl}(NK,\mathbb{R}), \tag{51}$$

where $B^{(h)}$ is the row-stochastic routing matrix of head h, with entries $\beta_{ij}^{(h)}$, and $M_h = W_O^{(h)}W_V^{(h)}$. With frames, the (i,j) block of $\mathfrak{L}_\beta$ is the affine-chart routed edge map $\mathcal{M}_{ij}$, covariant under $g = \bigoplus_i g_i$. Thus $I + \mathfrak{L}_\beta$ is the first-order chart of $\exp(\mathfrak{L}_\beta)$, whose identity is the residual connection; it is block lower triangular under causal routing and, on the principal-logarithm domain, the exponential of a real block lower triangular generator. At a single edge the layer is $\delta_{ij}I + O\boldsymbol{\beta}_{ij}T$, the affine chart with $c_{\mathrm{res}} = 1$, and neither

the tiled exponential message nor its first-order truncation (proofs and qualifications in Appendix B.5).

Whether $W_O W_V$ is invertible is thus beside the point: $W_O W_V = OT$ is a generator, and the group-valued objects are $e^{OT} \in \mathrm{GL}^+(K)$, always invertible, and $I + OT$, invertible exactly when -1 is not an eigenvalue of OT. The exponential chart replaces the output projection by $\widetilde{O} = e^{OT} T^{-1}$, confining the stacked transfer to $\exp(\mathfrak{gl}(K))$, with $O = 0$ giving the head-routed flat transport rather than zero content.

3.5 Holonomy on the causal graph

On the causal support $\mathcal{N}_i = \{0, \dots, i\}$ every closed directed walk is a self-loop (Theorem 11), but invertible links admit loop observables. For an invertible link Θ_{ij}, $j < i$, let $\mathring{\Theta}_{ij} = U_i^{-1} \Theta_{ij} U_j$.

Definition 8 (Triangle holonomy) *For $k < j < i$ put*

$$\mathcal{H}_{ijk} = \Theta_{ik}^{-1} \Theta_{ij} \Theta_{jk} = U_k \mathring{\Theta}_{ik}^{-1} \mathring{\Theta}_{ij} \mathring{\Theta}_{jk} U_k^{-1}. \tag{52}$$

The field is flat on the complete causal support when $\mathcal{H}_{ijk} = I$ for every strict triple.

Since $\mathcal{H}_{ijk} \mapsto g_k \mathcal{H}_{ijk} g_k^{-1}$ under the passive site action, its conjugacy class is invariant; it is the holonomy of the loop $k \to j \to i \to k$ based at k, whose last step traverses the long edge backward with its inverse. On graphs with missing edges a cycle set must be declared, and the recurrence characterization below need not apply.

Proposition 9 (Flatness and pure gauge)

1. *Let Θ_{ij}, $j < i$, be arbitrary $K \times K$ matrices with $\Theta_{ij}\Theta_{jk} = \Theta_{ik}$ for all $k < j < i$. Then $\Theta_{ij} = \Theta_{i,i-1}\Theta_{i-1,i-2} \cdots \Theta_{j+1,j}$, and conversely every choice of adjacent matrices defines a flat field through these ordered products.*

2. *If every Θ_{ij} is invertible, the field is flat exactly when $\Theta_{ij} = g_i g_j^{-1}$ for some $g_i \in \mathrm{GL}(K)$, and then $g_i = \Theta_{i0}$ up to a common right factor. For singular matrices this equivalence fails: a shared projector $\Theta_{ij} = P = P^2$ is flat and is not of the form $g_i g_j^{-1}$.*

The proof, by induction on $i - j$ and the cocycle identity, is in Appendix B.6.

Hence Ω_{ij} and every reference field $h_i h_j^{-1}$, such as a rotary link $e^{(p_i - p_j)Y}$, are pure gauge. Since $\mathcal{H}_{ijk}$ is conjugate to $\mathring{\Theta}_{ik}^{-1} \mathring{\Theta}_{ij} \mathring{\Theta}_{jk}$, no choice of frames creates or removes curvature, and a constant reference link $\mathring{\Theta}_{ij} = A \neq I$ is never flat.

With one shared generator, $\Theta_{ij} = \Theta_{ii}\Omega_{ij}$ with the self-link factor $\Theta_{ii} = U_i e^L U_i^{-1}$, and likewise

$$M_{ij}^{(h)} = M_{ii}^{(h)} \Omega_{ij}, \qquad M_{ii}^{(h)} = U_i \widetilde{O}_h T_h U_i^{-1}, \tag{53}$$

so $\mathcal{H}_{ijk} = U_k e^L U_k^{-1}$ on every triangle. The implemented field is flat only if $e^L = I$ and carries one conjugacy class per layer, that of e^L: it is group valued but composed of site data and one learned matrix. Edge-dependent generators give holonomy that varies between triangles, the latent-link model of Supplement S6, whose zero-width limit is the implemented chart (Remark S17); routed holonomies and curvature measures on $\mathrm{GL}(K)$ are in Supplement S3.

4 Gauge Covariance and the Reference-Coordinate Normal Form

The operator of Section 2.4 is written with frames, but every routing cost and message factors through the reference moments (ξ_i, C_i). This section makes that statement exact, identifies the parameter redundancies it implies, states what the covariance constrains, and gives the normal form showing where the learned frames act.

Lemma 10 (Invariance of divergences under invertible maps) *For probability laws P and Q on $\mathbb{R}^K$ with densities and any $A \in \mathrm{GL}(K)$,*

$$D_{\mathrm{KL}}(A_*P \| A_*Q) = D_{\mathrm{KL}}(P\|Q). \tag{54}$$

Theorem 11 (Reference-moment factorization) *Fix a layer, its comparison maps $S_{Q,h}$, $S_{K,h}$, its stored content maps T_h, O_h with effective receiving maps $\widetilde{O}_h$ in one chart of Eq. (43), its source priors, its temperatures, and its residual coefficient c_{res}, and write $M_h = \widetilde{O}_h T_h$ for the reference head transfers.*

1. *Every routing cost $D_{ij}^{(h)}$ and weight $\beta_{ij}^{(h)}$ depends on the states only through the reference moments (ξ_j, C_j), and the attention update has the reference form*

$$U_i^{-1}\mu_i^{\mathrm{att}} = c_{\mathrm{res}}\xi_i + \sum_h \widetilde{O}_h z_i^{(h)} = c_{\mathrm{res}}\xi_i + \sum_{j,h} \beta_{ij}^{(h)} M_h \xi_j, \tag{55}$$

   with the routed sent blocks $z_i^{(h)} = \sum_j \beta_{ij}^{(h)} T_h \xi_j$ of Eq. (28).

2. *Under the passive site action of Eq. (12), the routing weights are unchanged, and the head maps $M_{ij}^{(h)} = U_i M_h U_j^{-1}$, the routed edge maps $\mathcal{M}_{ij} = \sum_h \beta_{ij}^{(h)} M_{ij}^{(h)}$, and the messages transform as*

$$M_{ij}^{(h)\prime} = g_i M_{ij}^{(h)} g_j^{-1}, \qquad \mathcal{M}'_{ij} = g_i \mathcal{M}_{ij} g_j^{-1}, \qquad m'_i = g_i m_i. \tag{56}$$

3. *The operator is unchanged by the common reference change $(U_i, S_{Q,h}, S_{K,h}, T_h, O_h) \mapsto (U_i a, S_{Q,h}a, S_{K,h}a, T_h a, a^{-1}O_h)$ with $a \in \mathrm{GL}(K)$, under which the stored generator $L = OT$ transforms by the adjoint action $L \mapsto a^{-1}La$, by $(S_{Q,h}, S_{K,h}) \mapsto (AS_{Q,h}, AS_{K,h})$ with $A \in \mathrm{GL}(r_h)$, and by the headwise change $(T_h, O_h) \mapsto (B_h T_h, O_h B_h^{-1})$ with $B_h \in \mathrm{GL}(s_h)$, in each chart. The common reference change also leaves every operation in local coordinates unchanged.*

4. *For a closed walk $\gamma = (i_0, \dots, i_n = i_0)$, the routed loop product $\mathcal{H}_\gamma = \mathcal{M}_{i_0 i_1} \cdots \mathcal{M}_{i_{n-1} i_0}$ satisfies $\mathcal{H}'_\gamma = g_{i_0} \mathcal{H}_\gamma g_{i_0}^{-1}$, and its trace is a polynomial in the routing weights with loop-independent coefficients $\mathrm{Tr}(M_{h_1} \cdots M_{h_n})$. On the causal support $\mathcal{N}_i = \{0, \dots, i\}$, every closed walk is a self-loop.*

Appendix C proves the lemma, the theorem, and Corollary 12 below. Item (2) is a covariance under changes of each agent's coordinates; item (3) lists internal redundancies of the parameters, since the model determines $\mathcal{Y}_h$ and $\mathcal{Z}_h$ only as abstract vector spaces. The comparison redundancy is cogredient, unlike the contragredient redundancy of dot-product scores, and Appendix C.2 gives the balance laws that gradient flow conserves along both.

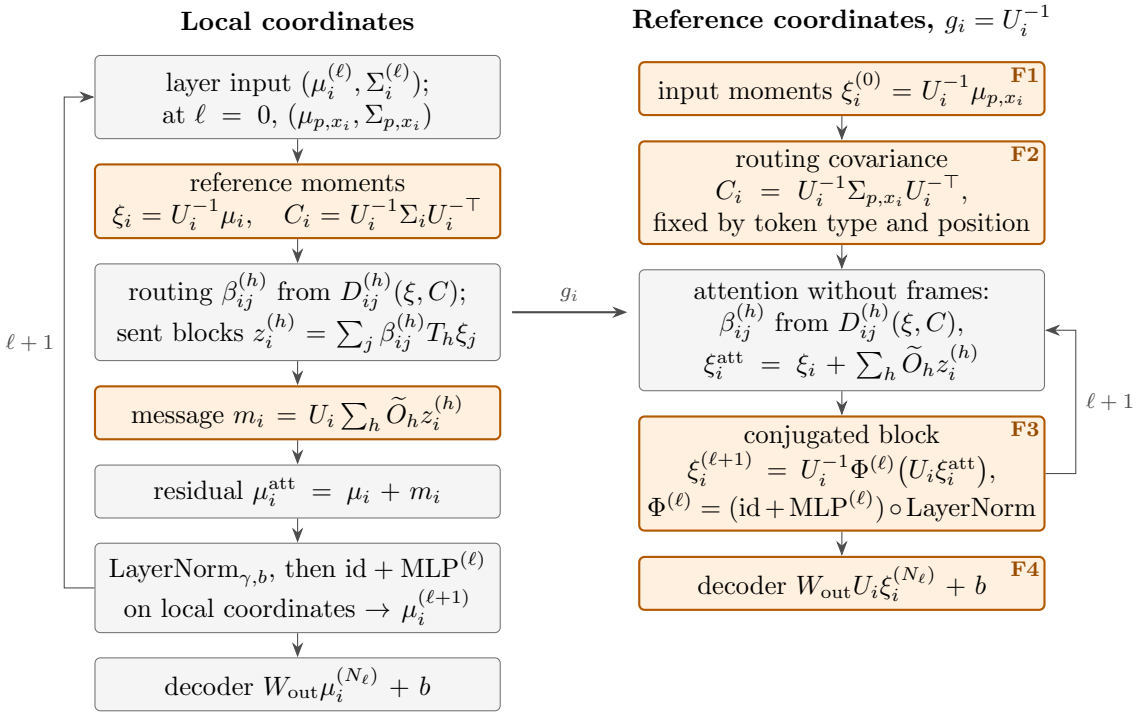


Figure 3: Reference-coordinate form of the evaluated architecture. Shaded boxes mark where the frames U_i enter; $z_i^{(h)}$ is the routed sent block of head h and $\widetilde{O}_h$ the effective receiving map of the chart. Left: one block in local coordinates. Right: the same computation after $g_i = U_i^{-1}$, where attention acts without frames (Eq. (56)) and the frames remain only in the input moments (F1), the routing covariance (F2), the conjugated normalization and MLP (F3), and the decoder (F4). Each tag names the row of Table 11 that removes that role. Under moment matching, C_i is fixed only at the input and then updated frame-free from the routing weights.

What the covariance constrains. The theorem uses only that routing and content read the states through (ξ_i, C_i), so $g_i = U_i^{-1}$ removes the frames, and any operator with this property, including dot-product attention on dressed means, is covariant in the same sense. In the terminology of Fournel et al. (2014), U_i is a local dressing field and the local $\mathrm{GL}(K)$ covariance is artificial (François, 2019, Sec. 2.2): the learned frames fix a global trivialization, an $\{e\}$-structure (Weiler et al., 2021, Sec. 2.1.1), and the covariance places no constraint on S_Q, S_K, T, O, or a link generator. The intertwiner requirement of gauge-equivariant networks (Section 1.1) would remove the low rank, block mixing, and composition defect of the selective links for $G = \prod_b \mathrm{GL}(d_b)$; these are available because the frames are learned (Supplement S7).

The evaluated architecture applies its normalization, MLP, and decoder in local coordinates, where the learned frames act beyond the attention operator.

Corollary 12 (Normal form of the evaluated predictor) *In reference coordinates the predictor of Section 2.4 reads*

$$\xi_i^{(0)} = U_i^{-1}\mu_{p,x_i}, \qquad C_i^{(0)} = U_i^{-1}\Sigma_{p,x_i}U_i^{-\top}, \qquad \xi_i^{(\ell+1)} = U_i^{-1}\Phi^{(\ell)}\big(U_i\xi_i^{\mathrm{att},(\ell)}\big), \tag{57}$$

with $\Phi^{(\ell)} = (\mathrm{id} + \mathrm{MLP}^{(\ell)}) \circ \mathrm{LN}$, *with the attention step frame-free given the routing weights, and with decoder logits* $W_{\mathrm{out}}U_i\xi_i^{(N_\ell)}$ *plus the class biases. Under covariance passthrough* $C_i^{(\ell)} = C_i^{(0)}$ *at every depth; under moment matching* $C_i^{(\ell+1)}$ *is the reference covariance of Eq.* (70)*, a frame-free function of the routing weights and the reference moments of layer* ℓ*.*

The frames therefore act through four channels (Figure 3): a multiplicative token-and-position embedding; a routing covariance, static per token type and position under passthrough and context dependent after the input under moment matching; a token- and position-specific conjugation of the shared normalization and MLP; and a token- and position-specific decoder $W_{\mathrm{out}}U_i$. The last two have covariant alternatives (Appendix C.3). In local coordinates, one gauge choice away, the frame enters only the attention operator, through $S_{Q,h}U_i^{-1}$, $S_{K,h}U_j^{-1}$, the projected covariances, and $U_i\widetilde{O}_hT_hU_j^{-1}$. Section 7.3 counts the frame parameters.

5 Gaussian-KL Routing and Its Transformer Reductions

This section writes the routing cost of Proposition 1 in dual coordinates, reduces it to biased dot-product attention, and gives the value/output dictionary (Table 2).

5.1 The full Gaussian score and its query–key cross term

Fix a layer, a head, and nonempty allowed source sets, and suppress layer and head indices except on r_h. Define the projected moments

$$a_i = S_Q\xi_i, \quad b_j = S_K\xi_j, \quad V_{Q,i} = S_QC_iS_Q^\top, \quad V_{K,j} = S_KC_jS_K^\top. \tag{58}$$

The state covariances are positive definite and both comparison maps have full row rank, so $V_{Q,i}, V_{K,j} \succ 0$. The Gaussian KL formula gives

$$\begin{aligned} D_{ij} = \frac{1}{2}\big[\,&\mathrm{Tr}(P_jV_{Q,i}) + a_i^\top P_ja_i - 2a_i^\top P_jb_j + b_j^\top P_jb_j - r_h \\ &+ \log\det V_{K,j} - \log\det V_{Q,i}\big], \qquad P_j = V_{K,j}^{-1}. \end{aligned} \tag{59}$$

The mean compatibility enters as $-a_i^\top P_jb_j$, in local coordinates $\mu_i^\top \mathsf{B}_{ij}\mu_j$ (Eq. (S11)), so independent S_Q, S_K supply distinct query and key observables. When P_j varies with the source, $a_i^\top P_ja_i$ and $\mathrm{Tr}(P_jV_{Q,i})$ are pair dependent, not row constants.

Theorem 13 (Dual-coordinate form of Gaussian-KL routing) *Write the query law in expectation coordinates and the key law in natural coordinates,*

$$\eta_{Q,i} = \big(a_i, V_{Q,i} + a_ia_i^\top\big), \qquad \theta_{K,j} = \big(P_jb_j, -\tfrac{1}{2}P_j\big), \qquad \langle\eta,\theta\rangle = \eta_1^\top\theta_1 + \mathrm{Tr}(\eta_2\theta_2), \tag{60}$$

Correspondence	Exact scope or remaining condition
Derived under stated conditions	
Gaussian categorical routing	Exact optimizer for fixed costs, $\tau > 0$; exact source posterior under a binary compatibility record (Theorem 15)
Query/key dot product	Exact dot product in dual coordinates (Theorem 13); mean block alone under shared key covariance
Ordinary attention weights	Shared key covariance, matched scale, constant or compensated key norms; otherwise whitened L2 attention
RoPE score	Relative dependence exact when both laws rotate; dot-product form needs a shared key covariance commuting with the rotations
Identical by construction	
Value/output aggregation	T_h and $\widetilde{O}_h$ are the value and output maps at any fixed routing
Exponential content link	$W_O = e^L T^{-1}$, $L = OT$, for tiled heads: W_O reparametrized on the real-logarithm domain
Multi-head organization	Concatenation identity; no frame-block partition needed
Token embedding	Prior mean μ_p^c is the embedding row; Σ_p^c affects means only via routing
Architectural choices, not derived or without counterpart	
Token and positional frames	Per-type map e^{ϕ_c}, no counterpart; positional map e^{Y_i} on queries, keys, values, and output, replacing additive positions or RoPE
Predictive residual	$c_{\text{res}} \in \{0, 1\}$, the persistence term of a predictive kernel; absent in tiled exponential aggregation
Covariance, normalization, MLP, decoder	Separate choices; the linear decoder is a Gibbs readout of shared-covariance templates (Appendix E)

Table 2: Transformer correspondences grouped by status; the equalities concern specified operators, not unrestricted model families.

and let $A(\theta_{K,j}) = \frac{1}{2} b_j^\top P_j b_j + \frac{1}{2} \log\det V_{K,j} + \frac{r_h}{2} \log 2\pi$ *be the Gaussian log-partition, with conjugate* $A^*(\eta_{Q,i}) = -H(q_{Q,i})$*. Then*

$$D_{ij} = A(\theta_{K,j}) + A^*(\eta_{Q,i}) - \langle \eta_{Q,i}, \theta_{K,j} \rangle, \tag{61}$$

and the categorical optimizer is exactly $\beta_{ij} = \mathrm{softmax}_j\big(\langle \eta_{Q,i}, \theta_{K,j} \rangle/\tau - A(\theta_{K,j})/\tau + \log \pi_{ij}\big)$*: a dot product in* $\mathbb{R}^{r_h + r_h(r_h+1)/2}$ *between query expectation parameters and key natural parameters, with the key log-partition as a source bias. No covariance hypothesis is needed.*

Proof With $-H(q_{Q,i}) = -\frac{1}{2}\log\det(2\pi e V_{Q,i})$, expanding the right side of Eq. (61) reproduces Eq. (59) term by term. This is the mixed form of the divergence between members of an exponential family (Wainwright and Jordan, 2008, Sec. 5.2, Eq. (5.11)). The term $A^*(\eta_{Q,i})$ is constant on row i and cancels in the softmax. ■

With a shared second-order key coordinate $-\frac{1}{2}P_j$, only the mean block $a_i^\top V_*^{-1} b_j$ of the pairing varies with the source.

Proposition 14 (Projected Gaussian reduction to biased dot-product routing) *Suppose all keys used by a head have the same projected covariance* $V_{K,j} = V_* \succ 0$*. Let* $\tau > 0$*, let each prior row be a probability distribution with* $\pi_{ij} > 0$ *on its allowed support, and let costs be the uncapped Gaussian divergences. Define* $\mathsf{q}_i = V_*^{-1/2} a_i$ *and* $\mathsf{k}_j = V_*^{-1/2} b_j$*, using the symmetric positive-definite inverse square root. Then the exact categorical optimizer is*

$$\beta_{ij} = \mathrm{softmax}_j\left(\frac{\mathsf{q}_i^\top \mathsf{k}_j}{\tau} - \frac{\|\mathsf{k}_j\|^2}{2\tau} + \log \pi_{ij} \right). \tag{62}$$

The query covariances may differ across receivers. If, more strongly, $V_{Q,i} = V_{K,j} = \sigma_h^2 I_{r_h}$*, then* $D_{ij} = \|a_i - b_j\|^2/(2\sigma_h^2)$*.*

Appendix C.4 gives the proof. Up to a row constant $D_{ij} = \frac{1}{2}\|\mathsf{q}_i - \mathsf{k}_j\|^2$, so this is L2 attention in whitened coordinates, under the isotropic hypothesis that of Kim et al. (2021). The query entropy is row constant, so the Gibbs row is the mean-field responsibility of a Gaussian mixture, for a point query that of Movellan and Gabbur (2020) and Nguyen et al. (2022) (Remark S10).

5.2 Key-norm bias and its compensation

Equation (62) is dot-product attention with the same prior exactly when the squared key norms are constant on the allowed support. A base prior π_{ij}^0 is reproduced by the state-dependent compensation

$$\pi_{ij} = \frac{\pi_{ij}^0 \exp(\|\mathsf{k}_j\|^2/(2\tau))}{\sum_{k \in \mathcal{N}_i} \pi_{ik}^0 \exp(\|\mathsf{k}_k\|^2/(2\tau))}, \tag{63}$$

which the evaluated model does not use. Otherwise the two routings differ: for $r_h = 1$, $\tau = 1$, $a_i = 0$, keys $b_1 = 0, b_2 = 1$, shared unit variance, and uniform prior, dot-product routing gives $(1/2, 1/2)$ and Gaussian routing $(1, e^{-1/2})/(1 + e^{-1/2})$. Neither LayerNorm nor high-dimensional concentration removes the key-norm term at $\tau = \sqrt{r_h}$; Supplement S4 bounds the routing error sharply (Proposition S9) and separates temperature from covariance scale.

5.3 Value, output, and residual links

Concatenating the $z_i^{(h)}$ of Eq. (55) gives the multi-head output projection $\widetilde{O}\,\mathrm{concat}_h(z_i^{(h)})$ (Vaswani et al., 2017, Sec. 3.2.2) with $W_V^{(h)} = T_h$ and $W_O^{(h)} = \widetilde{O}_h$ in column-vector convention, free in the affine chart and, for tiled heads in the exponential chart, restricted to $W_O = e^L W_V^{-1}$ on the real-logarithm domain (Theorem 7). In local coordinates the operation is $\sum_{j,h} \beta_{ij}^{(h)} M_{ij}^{(h)} \mu_j$, so $M_{ij}^{(h)}$ is the framed value/output composition, not the query–key tensor, and the residual $c_{\mathrm{res}} = 1$ is the persistence term of the predictive kernel of Eq. (69). Rotating both projected laws, as rotary embedding rotates queries and keys (Su et al., 2024), gives by Lemma 10 a relative-position score for arbitrary covariances (Supplement S4.4).

6 Probabilistic Status of Routing and Content

This section shows that the Gibbs row of Eq. (24) is an exact posterior of a declared local source model at every temperature and that the residual content update of Eq. (28) is an exact predictive mean.

6.1 Variational source selection

For one layer, head, and receiver (indices suppressed), let $q_{Q,i}(y)$ be the projected query law, $p_{K,j}(y)$ the candidate key laws, and $\pi_{ij} > 0$ prior probabilities summing to one on a nonempty allowed support $\mathcal{N}_i$. The local source model and factorized recognition law are

$$\mathbb{P}_i(y, J = j) = \pi_{ij} p_{K,j}(y), \tag{64}$$

$$\mathbb{Q}_{i,\beta}(y, J = j) = \beta_{ij} q_{Q,i}(y). \tag{65}$$

The states enter as data fixed by the prefix before routing; the query law is the clamped recognition marginal of y, and β_i is the only recognition variable. Expanding the density ratio gives

$$D_{\mathrm{KL}}(\mathbb{Q}_{i,\beta} \| \mathbb{P}_i) = \sum_{j \in \mathcal{N}_i} \beta_{ij} D_{ij} + \sum_{j \in \mathcal{N}_i} \beta_{ij} \log \frac{\beta_{ij}}{\pi_{ij}}, \tag{66}$$

$$D_{ij} = D_{\mathrm{KL}}(q_{Q,i} \| p_{K,j}), \tag{67}$$

so the categorical entropy belongs to the variational objective. Key laws as generative components and the query law as recognition factor fix the direction of D_{ij}; Supplement S4 records the other directions. At a declared temperature $\tau > 0$ the entropy-regularized objective has the unique interior minimizer over the simplex (Appendix C.5)

$$\beta_{ij}^* = \frac{\pi_{ij} \exp(-D_{ij}/\tau)}{\sum_{k \in \mathcal{N}_i} \pi_{ik} \exp(-D_{ik}/\tau)}, \tag{68}$$

which is the Gibbs row of Eq. (24). The objective is the KL of the declared joint at $\tau = 1$ and an entropy-regularized extension otherwise; Theorem 15 supplies a normalized model in which the row is exact at every temperature.

Complete-state alignment, $S_Q = S_K = I_K$, scores sender j by $D_{\mathrm{KL}}(q_i \| (\Omega_{ij})_* q_j)$, and the comparison link is the content link Θ_{ij} exactly when $S_K = S_Q e^L$ (Appendix C.5); the evaluated model imposes neither.

6.2 Exact posterior and predictive mean

Theorem 15 (Exact probabilistic status of routing and content) *Fix a layer and its input states, and let $D_{ij}^{(h)} \geq 0$ be the routing costs.*

1. *For every $\tau_h > 0$, give each receiver and head a source label $J_i^{(h)}$ with prior $\pi_i^{(h)}$ and a binary compatibility record $o_i^{(h)}$ with $P(o_i^{(h)} = 1 \mid J_i^{(h)} = j) = \exp(-D_{ij}^{(h)}/\tau_h)$. The joint is normalized on any source graph, and $P(J_i^{(h)} = j \mid o_i^{(h)} = 1) = \beta_{ij}^{(h)}$ of Eq. (24). The objective of Eq. (23) is τ_h times the label free energy $D_{\mathrm{KL}}(\beta \| \pi_i^{(h)}) - \sum_j \beta_j \log P(o_i^{(h)} = 1 \mid J_i^{(h)} = j)$, whose minimum is the surprisal $-\log P(o_i^{(h)} = 1)$.*

2. *Let the head labels be independent with laws $\beta_i^{(h)}$, and take the directed kernel*

$$k_i' \mid k, J_i \sim \mathcal{N}\Big(c_{\mathrm{res}} k_i + \sum_h M_{iJ_i^{(h)}}^{(h)} k_{J_i^{(h)}}, Q_i\Big), \qquad c_{\mathrm{res}} \in \{0, 1\}, \quad Q_i \succeq 0, \tag{69}$$

which is normalized for any rank of the head maps, as a degenerate Gaussian when Q_i is singular. At state values $k_j = \mu_j$ its predictive mean is μ_i^{att} of Eq. (28). With independent source states $k_j \sim \mathcal{N}(\mu_j, \Sigma_j)$, the self edge included, its moment-matched covariance in reference coordinates is

$$\begin{aligned} C_i^{\mathrm{att}} = \widetilde{Q}_i + c_{\mathrm{res}}\Big[C_i + \sum_h \beta_{ii}^{(h)}\big(M_h C_i + C_i M_h^\top\big)\Big] + \sum_j \sum_{h,h'} c_{ij}^{hh'} M_h C_j M_{h'}^\top \\ + \sum_h \Big[\sum_j \beta_{ij}^{(h)} n_{ij}^{(h)} n_{ij}^{(h)\top} - \bar{n}_i^{(h)} \bar{n}_i^{(h)\top}\Big], \end{aligned} \tag{70}$$

where $\widetilde{Q}_i = U_i^{-1} Q_i U_i^{-\top}$, $M_h = \widetilde{O}_h T_h$, $c_{ij}^{hh} = \beta_{ij}^{(h)}$, $c_{ij}^{hh'} = \beta_{ij}^{(h)} \beta_{ij}^{(h')}$ for $h \neq h'$, $n_{ij}^{(h)} = M_h \xi_j$, and $\bar{n}_i^{(h)} = \sum_j \beta_{ij}^{(h)} n_{ij}^{(h)}$. The local covariance $U_i C_i^{\mathrm{att}} U_i^\top$ obeys the passive law.

Proof [Proof of item 1] Because $D_{ij}^{(h)} \geq 0$, each record kernel takes values in $(0, 1]$ and is normalized. Records attached to different receivers and heads are conditionally independent children of the states and labels, so the joint is normalized. Bayes' rule gives $P(J = j \mid o = 1) = \pi_{ij} e^{-D_{ij}/\tau} / \sum_k \pi_{ik} e^{-D_{ik}/\tau}$, which is Eq. (24). Since $-\log P(o = 1 \mid J = j) = D_{ij}/\tau$, the label free energy equals τ^{-1} times the objective, and its minimum is $-\log \sum_j \pi_{ij} e^{-D_{ij}/\tau} = -\log P(o = 1)$. Item 2 is proved in Appendix C.5. ∎

The record is a declared modeling event, as for optimality variables in control as inference (Toussaint and Storkey, 2006; Kappen et al., 2012; Levine, 2018), not an observed datum; since such exactness holds for any nonnegative cost, item 1 shows that Gibbs routing is consistent with exact inference, not that attention must be inference. Item 2 holds in every chart of Eq. (43) with T_h, $\widetilde{O}_h$, and Q_i free; it declares the persistence coefficient, additivity across heads, and one label for record and content, and excludes normalization, the MLP, and the collapse of the predictive mixture to a point state between layers. Appendix C.5 gives the covariance floor; Supplement S8 relates it to a normalized population-level latent-source model.

6.3 Receiver inference and latent links

Read as evidence about the receiver, the routed messages give a Gaussian power posterior whose closed-form minimizer fuses a receiver prior with the messages by a Kalman-type gain and tends to the residual-free head sum of Eq. (28) in a stated limit (Supplement S5). Promoted to latent variables with a normalized prior anchored at $L = OT$, edge- and depth-dependent generators L_{ij}^{ℓ} give an unimplemented model whose zero-width limit is the exponential chart and whose MAP stationarity condition is a link field equation, linearized for one head around a flat anchor to a screened, massive discrete Maxwell equation on the depth-by-token complex (Supplement S6).

7 Language-Model Realization

The evaluated model realizes the causal architecture of Sections 2.4 and 2.4 at $K = 128$ with four layers, the exponential chart of Eq. (43), moment-matched covariance, and learned positional frames, trained on WikiText-103 (Merity et al., 2017); its saved configuration and protocol define the empirical claim. Section 7.3 sets it beside protocol-matched baselines, and Section 7.5 compares the charts at $K = 20$. An affine-chart run at $K = 160$ reaches the lowest test perplexity reported here but forms no group-valued link; it and an affine run at $K = 256$ are in Supplement S10.

7.1 Forward computation

Algorithm 1 specifies the evaluated run; each layer performs one predictive readout and no iterative belief update. With $H = B = 4$ heads and $d_b = r_h = s_h = 32$, the stacked $T^{(\ell)}, O^{(\ell)} \in \mathbb{R}^{K\times K}$ define the shared generator $L^{(\ell)} = O^{(\ell)}T^{(\ell)}$ and the applied receiving map

$$\widetilde{O}^{(\ell)} = \big(T^{(\ell)}\big)^{-1} + O^{(\ell)}\varphi_1\big(T^{(\ell)}O^{(\ell)}\big) = e^{L^{(\ell)}}\big(T^{(\ell)}\big)^{-1}, \tag{71}$$

so $\widetilde{O}^{(\ell)}T^{(\ell)} = e^{L^{(\ell)}}$, and an edge on which every head gives the same source the same weight carries that weight times $\Theta_{ij} = U_i e^{L^{(\ell)}} U_j^{-1}$ (Theorem 7). The source prior is uniform on the causal support $j \le i$, so positions enter only through the positional frames. Every KL cost is bounded to $[0, 8K] = [0, 1024]$, which keeps the routing the exact categorical optimum and exact label posterior of Theorem 15, the bounded cost being nonnegative, and equal to the unbounded routing when the bound is inactive (Appendix F.3).

7.2 Architecture and training of the evaluated run

The evaluated run has $K = 128$, four frame blocks of dimension 32, $N_\ell = 4$ layers, and model seed 6, with per-layer maps $S_{Q,h}, S_{K,h}, T_h \in \mathbb{R}^{32\times 128}$ and $O_h \in \mathbb{R}^{128\times 32}$ initialized so that $\widetilde{O}T = I$ (Appendix F.2). Routing compares full 32×32 projected covariances with $\tau_h = \kappa_\beta\sqrt{d_b} = \sqrt{32}$ and $\kappa_\beta = 1$, with no ALiBi bias (Press et al., 2022) or rotary rotation. Each of the 50,257 vocabulary items and 128 positions has learned block-diagonal frame generators in $\mathfrak{gl}(32)^4$, so $\Omega_{ij} = U_i U_j^{-1}$ carries no holonomy. Covariance is moment matched and stored diagonally, a partial gauge fixing (Appendix F.1), and the normalization and MLP pass it through. Each layer applies one shared affine LayerNorm (Ba et al., 2016) to $\mu_i + m_i$ and a residual GELU MLP (Hendrycks and Gimpel, 2016) of hidden width 1024,

Algorithm 1 Causal Gaussian attention with the exponential content chart, as evaluated

Require: Tokens $x_0, \dots, x_{N-1}$ and, in training, targets $y_i = x_{i+1}$; the prior bank $\{(\mu_p^c, \Sigma_p^c, \phi_c)\}_{c=1}^V$ with diagonal Σ_p^c; positional generators $Y_0, \dots, Y_{N-1}$; layer maps $S_{Q,h}^{(\ell)}, S_{K,h}^{(\ell)}, T_h^{(\ell)}, O_h^{(\ell)}$; decoder parameters.

Initialize $(\mu_i^{(0)}, \Sigma_i^{(0)}) = (\mu_p^{x_i}, \Sigma_p^{x_i})$ and realize the frame $U_i = \exp(\phi_{x_i}) \exp(Y_i)$ once for all layers.
Form each $\widetilde{O}^{(\ell)}$ of Eq. (71) in double precision and split it into head blocks $\widetilde{O}_h^{(\ell)}$.
Set $\log \pi_{ij} = 0$ for $j \le i$ and $\pi_{ij} = 0$ otherwise.
for $\ell = 0, \dots, N_\ell - 1$ **do**
  Form $\xi_i = U_i^{-1} \mu_i^{(\ell)}$ and $C_i = U_i^{-1} \Sigma_i^{(\ell)} U_i^{-\top}$.
  for $h = 1, \dots, H$ **do**
    Form the query and key Gaussian pushforwards under $S_{Q,h}^{(\ell)}$ and $S_{K,h}^{(\ell)}$ with full reduced covariances; denote their KL divergence by $D_{ij}^{(\ell,h)}$.
    Bound the cost, $\bar{D}_{ij}^{(\ell,h)} = \min\{\max\{D_{ij}^{(\ell,h)}, 0\}, 8K\}$, and set $\beta_{ij}^{(\ell,h)} = \mathrm{softmax}_j\big(\log \pi_{ij} - \bar{D}_{ij}^{(\ell,h)}/\tau_h\big)$ with $\tau_h = \sqrt{r_h}$.
    Compute $z_i^{(\ell,h)} = \sum_j \beta_{ij}^{(\ell,h)} T_h^{(\ell)} \xi_j$.
  end for
  Compute $m_i^{(\ell)} = U_i \sum_h \widetilde{O}_h^{(\ell)} z_i^{(\ell,h)}$ and $\mu_i^{\mathrm{att},(\ell)} = c_{\mathrm{res}} \mu_i^{(\ell)} + m_i^{(\ell)}$ with $c_{\mathrm{res}} = 1$.
  Set $\Sigma_i^{(\ell+1)}$ to the moment-matched covariance of Eq. (70) with the reference floor $10^{-4} \operatorname{Tr}(C_i) I/K$, mapped to local coordinates and projected onto the local diagonal by matching marginal variances.
  Set $\bar{\mu}_i^{(\ell)} = \mathrm{LayerNorm}_{\gamma,b}(\mu_i^{\mathrm{att},(\ell)})$.
  Set $\mu_i^{(\ell+1)} = \bar{\mu}_i^{(\ell)} + W_2^{(\ell)} \,\mathrm{GELU}(W_1^{(\ell)} \bar{\mu}_i^{(\ell)} + b_1^{(\ell)}) + b_2^{(\ell)}$; the normalization and MLP leave $\Sigma_i^{(\ell+1)}$ unchanged.
end for
Compute $p(y_i \mid x_{\le i}) = \mathrm{softmax}\big(W_{\mathrm{out}} \mu_i^{(N_\ell)} + b_{\mathrm{out}} + \kappa_{\mathrm{uni}} \log p_{\mathrm{uni}}\big)_{y_i}$.
if training **then**
  Backpropagate the mean next-token cross-entropy to every trainable parameter and update with AdamW (Loshchilov and Hutter, 2019).
end if

with no final normalization, and the decoder of Eq. (36) is untied from the prior means, with $\kappa_{\mathrm{uni}} = 1$ (Appendix F.1). The residual coefficient is $c_{\mathrm{res}} = 1$, which with the exponential chart counts the receiver twice, through $\beta_{ii}^{(h)}$ and through the residual (Appendix B.5); the evaluated layer is thus an architectural choice that contains the exponential group link, not a derived update.

Training used 150,000 AdamW updates with batch 32 at context 128, 614,400,000 training-token presentations over 116,840,318 unique tokens (5.26 passes), and data seed 3, with no skipped or nonfinite step. Evaluation uses context 128 and stride 32 and scores each of the 280,541 test targets once under tiktoken GPT-2 tokenization ($V = 50{,}257$), so

the reported perplexity is the log-loss of a normalized windowed sequence law, token level and not comparable to word-level WikiText-103 results.

7.3 Correspondence with a conventional transformer and parameter accounting

Read as a decoder-only transformer (Table 2), the prior mean μ_p^c is the token embedding and the decoder the unembedding, untied and with the same unigram prior in the realization and its baselines. $S_{Q,h}, S_{K,h}, T_h, O_h$ have the shapes of $W_Q^{(h)}, W_K^{(h)}, W_V^{(h)}, W_O^{(h)}$, giving $4K^2$ attention parameters per layer, and the feed-forward block of width $8K$ is the same. With identity frames and a common isotropic covariance the model is a width-K transformer with distance-based attention plus a per-key bias (Section 5) and one shared normalization.

The per-type and per-position storage has no transformer counterpart: each vocabulary entry also stores a diagonal variance and a block-diagonal frame generator with $\sum_b d_b^2 = Kd_b$ entries, and each position a generator of the same size. The frame enters every layer, also through the content map $M_{ij}^{(h)} = U_i \widetilde{O}_h T_h U_j^{-1}$, which depends on the type and position of sender and receiver, where a transformer applies one $W_O^{(h)} W_V^{(h)}$ to every pair (Section 4). Thus K sets the shapes that d_{model} sets in a transformer, while each token type stores $K(d_b + 2) = 34K$ numbers, mostly as a matrix that multiplies the state in every layer: a matrix-space word representation (Rudolph and Giesbrecht, 2010; Socher et al., 2012; Mai et al., 2019) acting as a multiplicative token-conditioned map (Jayakumar et al., 2020), as in a hypernetwork (Ha et al., 2017). The positional frames extend rotary query-key rotations (Su et al., 2024) to invertible blocks that also act on content.

The evaluated run has 227,041,490 parameters. The token-frame table holds $50{,}257 \times 4 \times 32^2 = 205{,}852{,}672$ of them (90.67%) and the vocabulary-indexed tables 225,201,617 (99.19%), while the context-processing parameters number 1,315,584 (Appendix F.4). Under the convention of Kaplan et al. (2020), which counts only non-embedding parameters, the realization at $K = 128$ and the transformer of width 128 both have 1.32M parameters and differ in the input tables, 218.7M against 6.4M (Table 3). That convention presumes additive embeddings, whereas the frame table computes in every layer, so the per-type table must be reported with every perplexity; the count measures capacity, not compute.

Within each of the first two groups of Table 3, the realization and the transformer share tokenizer, evaluation protocol, depth, heads, feed-forward expansion, batch, 30,000 or 15,000 updates, 122.9M training tokens (a little over one pass), and non-embedding count; both realizations predate the chart selector and use the affine chart. They differ in total parameters (factors of 16 and 17), in optimizer settings (tuned for the realization, a default with unswept learning rate for the baselines), in normalization placement and positional mechanism, and in training time per token (larger for the realization on the same device by about eight times at width 128 and five times at width 64). At matched tokens the realization reaches 39.77 against 90.15 at width 128 and 52.91 against 117.63 at width 64. The evaluated run shares the architecture and evaluation protocol of the realization at $K = 128$, but saw five times the tokens and differs in the chart, the source prior, the covariance contract, and the settings listed in Appendix F.2, so 34.48 against 39.77 attributes the difference to no single change. The best affine-chart run, 30.14, uses

Model	Chart	K	N_ℓ	Context/ stride	Tokens	Input table (per type)	Non-emb.	Total	Test PPL
Evaluated run	exp.	128	4	128/32	614.4M	218.7M (4,352)	1.32M	227.0M	34.48
Realization	affine	128	4	128/32	122.9M	218.7M (4,352)	1.32M	227.0M	39.77
Transformer	–	128	4	128/32	122.9M	6.4M (128)	1.32M	14.2M	90.15
Realization	affine	64	4	128/32	122.9M	109.4M (2,176)	0.33M	113.2M	52.91
Transformer	–	64	4	128/32	122.9M	3.2M (64)	0.33M	6.8M	117.63
Best affine run	affine	160	6	256/128	491.5M	273.4M (5,440)	3.08M	285.9M	30.14

Table 3: Parameter accounting and protocol. Input table: prior means, log-variances, and frame generators per vocabulary entry ($V = 50{,}257$), or a transformer's token embedding (numbers per entry in parentheses). Non-emb.: attention, feed-forward, and normalization. Total adds the decoders (6.48M, 3.27M, 8.09M at $K = 128$, 64, 160, equal at matched width) and the realizations' positional frames (0.52M, 0.26M, 1.31M). Tokens: training-token presentations (training set 116.8M). Context/stride: training context, evaluation stride. Priors: uniform causal (evaluated run), ALiBi with learned positional frames (other realizations), ALiBi alone (transformers). The Realization and Transformer rows of each of the first two groups share one protocol. Every row scores the 280,541 test targets once, with one seed.

Evaluation	Step	CE (nats)	Perplexity
Best recorded validation	140,000	3.4948	32.94
Final validation	150,000	3.5098	33.44
Test, final weights	150,000	3.5404	34.48
Test, best-validation weights	140,000	3.5255	33.97
Training sample, evaluation protocol, final weights	150,000	3.2095	24.77

Table 4: Recorded evaluations of the evaluated run ($K = 128$, four layers, exponential chart, one seed); the training sample is described in Appendix F.6.

context 256 with stride 128, six layers, width 160, and four times the tokens over 4.2 passes, and has no matched baseline.

7.4 Results and trained links

Validation cross-entropy (Table 4) reached its minimum at step 140,000 and rose by 0.015 nats by the final step while the training-sample cross-entropy kept falling; the test cross-entropy at the final weights, giving the headline 34.48, exceeds that at the best-validation weights by the same amount. The run supplies no seed variability, full-split intervention, or retraining ablation; Appendix F reserves the matched comparison, including $c_{\mathrm{res}} = 0$ (Table 7), and the frozen interventions (Table 11).

Trained exponential links. The trained links lack the conditioning of the small-width exponential arms of Section 7.5. In double precision from the final weights (Appendix F.5), every layer has $\det e^{L^{(\ell)}} = e^{\operatorname{tr} L^{(\ell)}} \le e^{-674}$ and an eigenvalue of $L^{(\ell)}$ with real part at most -39.5, so $\sigma_{\min}(e^{L^{(\ell)}}) \le e^{-39.5} \approx 7 \times 10^{-18}$ because $\sigma_{\min}(A) \le \min|\lambda(A)|$, and the numerical rank of $e^{L^{(\ell)}}$ at relative threshold 10^{-3} is 47, 61, 83, and 90 of 128. The group link is invertible in exact arithmetic (Proposition 3) but numerically of low rank, a selective channel reached by contracting most directions rather than by the rank deficiency of a single head. Its inverse has spectral norm at least e^{123} in one layer, since $\|A^{-1}\|_2 \ge 1/\min|\lambda(A)|$, so holonomy and inverse-edge diagnostics must be computed in log space (Supplement S3). These describe one single-seed checkpoint, not a trend in width or training length.

7.5 Link charts and the channel receiver at small width

Four runs compare the three charts of Eq. (43) and the channel receiver of Supplement S5 at $K = 20$ with two heads of width 10, two layers, $c_{\text{res}} = 0$, 15,000 updates, and one seed and data order; every arm starts from $\widetilde{O} = I$ and is evaluated as in Section 7.2 (Appendix F.8).

Arm	Content update	Test CE	Test PPL	Validation PPL	Time (s)
Affine, $\widetilde{O} = O$	predictive mean	4.80216	121.77	121.41	1344
First order, $\widetilde{O} = T^{-1} + O$	predictive mean	4.80390	121.98	122.38	1372
Exponential, $\widetilde{O} = e^{O^T} T^{-1}$	predictive mean	4.80016	121.53	121.69	1464
Exponential, $\widetilde{O} = e^{O^T} T^{-1}$	channel receiver	4.82773	124.93	124.98	1677

Table 5: Matched single-seed runs at $K = 20$ with two layers. Cross-entropy (CE) is in nats per scored token and perplexity (PPL) is its exponential; validation values are at the final step, the best recorded validation step in every arm; time is wall-clock training time on one RTX 5090. The receiver arm also holds the covariance at its encoding value instead of moment matching it.

The three predictive arms differ by at most 0.0037 nats per token on test and 0.0080 on validation, and their order differs between the splits. The largest test difference lies below the standard deviation of a two-run difference, about 0.0052 nats, implied by the test cross-entropy standard deviation of 0.0037 nats (0.33 perplexity points near 87) over three seeds of a different configuration at $K = 30$ (four layers, two heads of width 15, exponential chart, learned positional frame), whose own 95% interval is 0.17 to 2.1 perplexity points; the comparison fixes an order of magnitude, not a significance level. Equal held-out loss is what comparable optimization predicts, since the three charts represent the same tiled mean maps up to the real-logarithm domain (Proposition 17 and Theorem 7); the comparison probes optimization geometry and domain, not expressivity.

[**Data pending:** paired replicate seeds of all four arms, with seed-to-seed standard deviations measured on these configurations.]

The trained links differ as the charts predict: the first-order transfer crossed the singular set, while the exponential transfers stayed positive-determinant and better conditioned (Supplement S10.3). The receiver arm is 0.0276 nats worse on test than the exponential

predictive arm but also differs in the covariance contract, so the difference does not isolate the receiver.

8 Discussion

8.1 Contribution and limits

The contribution is a construction in which token occurrences are statistical agents over one context and attention is their communication, with the exact identities of Theorems 7, 11, 13, and 15, checked in float64 at small sizes (Supplement S9). No priority is claimed for separate comparison and content maps or for Gaussian-KL or gauge-equivariant routing; the construction supplies the setting in which their roles are defined. The independent-content joint gives the attention sublayer a probabilistic reading but leaves the content maps underdetermined. With one shared generator per layer, the exponential chart changes the parametrization, center, and domain of the content maps but not the family of tiled mean maps (Proposition 17); whether group geometry helps optimization at larger width, or becomes expressive once edge-dependent generators carry an action (Supplement S6), is open. The comparisons of Section 7 establish no parameter or sample efficiency, no benefit from learned frames or a chart, and no necessity of any component.

8.2 Toward non-neural language architectures

The evaluated model is a hybrid. Its affine LayerNorm, coordinate GELU MLP, and linear decoder are standard network components, and every parameter is trained by backpropagation of the next-token cross-entropy. The communication layer is stated without neurons: its states are beliefs, its frames private coordinates, its interface maps a protocol shared within a layer, its routing weights a source posterior, its content a predictive mean, and its transport a group-valued link. That layer is the non-neural candidate, and Section 7 shows that it trains inside a working predictor when the neural components are kept. Three pieces are missing before the working hypothesis can be tested. The MLP acts on each agent's mean with shared weights, a private computation rather than communication, but no agent-level interpretation of it is derived; that SiLU and GELU are exponential self-links (Remark S4) derives nothing further. Predictive-coding networks approximate backpropagation by local variational updates (Whittington and Bogacz, 2017; Millidge et al., 2022), but whether these agents can learn their priors, frames, and interface maps by local inference is untested. No joint law of the stack makes layer composition descend one free energy; the evidential receiver (Supplement S5) acts within one layer. Reading the links as parallel transports along a discretized base would also require agents at distinct contexts (Appendix A). Tying the key map to the transmitted content and absorbing the message by Bayesian conditioning would derive the receiving map and the residual rather than learn them; this variant is not developed here.

8.3 Experiments needed to separate the contributions

A conventional transformer with a learned per-type block read map of the frame table's size, and realizations with frames frozen at initialization or set to the identity, would separate frame geometry from table size; both controls are implemented and unrun. The conventional

baselines need a learning-rate sweep selected on validation, and isolating covariance-aware routing needs a mean-only or dot-product retraining control, which the current code no longer provides. The exponential chart should be trained with $c_{\text{res}} = 0$ (Section 3) at the protocol of the affine run at $K = 160$ (Supplement S10.1). Every arm needs replicates varying both model and data-order seeds, and trained-link spectra and holonomies should be recorded across training in log space (Section 7.4). Tables 11 and 7 complete this program.

9 Conclusion

This paper builds the communication layer of a program that reads neural language models as one realization of interacting statistical agents. The layer is exact where it is stated, trains inside a language model, and leaves open the matched comparison of Section 8.3 and the non-neural completion of Section 8.2. [**Data pending:** replace with the final matched comparison of Table 7 and the frozen interventions of Table 11.]

Code availability

The evaluated route is implemented in VFE 3.0 (`https://github.com/cdenn016/VFE_3.0`), in which the content-link chart is selected by `content_link_chart`. The run of Section 7.2 used `content_link_chart=exponential` and is stored in the run directory `34.48_wikitext-103_K128_block_glk_linear_mlp_s6` with its configuration, summary, and provenance records; Appendix F states their provenance limits. The channel receiver of Supplement S5 is selected by `attention_mode=projected_gradient` with `relational_receiver_law=channel_likelihood` and `relational_receiver_update=closed_form`, and each run of Section 7.5 saved an archive of its executed source. The per-type-table and frozen-frame controls of Section 8.3 are implemented in the same repository. A tagged release containing the evaluated route, a restored mean-only and dot-product control route, and the configurations of the reported runs is required. [**Data pending:** tag, commit, and archive identifier of that release.] The two affine-chart runs of Supplement S10 were executed from dirty source trees of the historical reference implementation (`https://github.com/cdenn016/V3_Transformer`). The mean-only and dot-product control route was removed from that repository after those runs, the current configuration loader rejects the archived configuration of the run at $K = 256$, and a public repository does not by itself recover the modified source bytes used in either run. The float64 identity checks of Supplement S9 and the measurement script for Table S4 are distributed with the manuscript source.

Acknowledgments and Disclosure of Funding

The author used Claude Opus 4.5, Claude Opus 4.6, and Claude Opus 5.5 (Anthropic) and OpenAI Codex for programming, figures, LaTeX, code inspection, and manuscript editing. The author remains responsible for the claims and computational record. The author received no funding and declares no conflicts of interest.

Appendix A. Bundle Details

This appendix records the bundle-theoretic details behind Section 2.2. For token attention the contextual dependence is evaluated at a fixed context c_*, while $i \in \mathcal{I} = \{0, \dots, N-1\}$ labels the token agents. Keeping a separately labeled copy of the fiber for each token is the pullback along the constant map $\iota : \mathcal{I} \to \mathcal{C}$, $\iota(i) = c_*$:

$$\mathcal{P}_{\mathcal{I}} = \iota^* \mathcal{P}, \qquad E_{\mathcal{I}} = \iota^* E. \tag{72}$$

Write E_i for the vector fiber at token i. This notation distinguishes token incidence from the contextual base of the general construction. Because ι is constant, $\iota^*\mathcal{P} = \mathcal{I} \times \mathcal{P}_{c_*}$ canonically, so the token construction sees no connection, curvature, or topology of $\mathcal{P}$. A vertical automorphism of $\iota^*\mathcal{P}$ is an independent automorphism of each copy of the G-torsor $\mathcal{P}_{c_*}$, and once a reference frame is fixed each such automorphism is one element $g_i \in G$; the gauge group of the token construction is therefore G^N. A gauge transformation of the contextual bundle restricts at c_* to a single automorphism of $\mathcal{P}_{c_*}$ and acts on the tokens only through the diagonal of G^N. The independent site changes of Eq. (12) are thus gauge transformations of the pulled-back bundle that the contextual bundle induces only when all g_i coincide, and they act as passive changes of the agents' coordinates.

The section u_i of each token agent and the reference trivialization give the frame $U_i : \mathsf{V} \to E_i$, and the statistical fiber of agent i is

$$\mathcal{G}(E_i) = \{\mathcal{N}(\mu, \Sigma) : \mu \in E_i,\ \Sigma \succ 0\}, \tag{73}$$

with the Fisher metric of Eq. (7) and the group action of Eq. (4).

A general contextual principal bundle need not admit a global section, and the factorized attention formulas use the reference trivialization σ_0 only on their working domain. Over the point c_* a reference frame always exists, since $\mathcal{P}_{c_*}$ is a nonempty G-torsor, and any two choices differ by right multiplication by one element of G. Replacing σ_0 by $\sigma_0 \cdot a$ with u_i held fixed gives $U_i \mapsto U_i a$, the common reference change of Section 2.2, with one $a \in \mathrm{GL}(K)$ for all agents. It changes the reference moments as $\xi_i \mapsto a^{-1}\xi_i$ and $C_i \mapsto a^{-1} C_i a^{-\top}$, leaves the local moments unchanged, and changes the representations of the shared interface maps of Section 2.3 as $S_{Q,h} \mapsto S_{Q,h}a$, $S_{K,h} \mapsto S_{K,h}a$, $T_h \mapsto T_h a$, and $O_h \mapsto a^{-1}O_h$. In each chart of Section 3 the applied receiving map then transforms as $\widetilde{O}_h \mapsto a^{-1}\widetilde{O}_h$ and a generator by the adjoint action $L \mapsto a^{-1}La$, so the measured laws and the content maps between agents are unchanged, and the operator is unchanged (Theorem 11). Holding the interface matrices fixed while changing the reference basis would specify a different communication rule.

The communication graph is additional data. Through the canonical identification of the copies of the fiber over c_*, the flat identification $\Omega_{ij} = U_i U_j^{-1}$ is the transition function between the sections u_j and u_i: it converts the u_j-coordinates of a vector into its u_i-coordinates, which is why it is pure gauge (Section 3). A group link with a nonzero generator, and every singular head map, is additional data on the edges of the graph rather than a transition function of $\mathcal{P}$ or the parallel transport of a connection on the base. Placing tokens at distinct contexts $\iota(i) = c_i$ and realizing links by parallel transport along declared base curves would make the contextual connection enter the attention operator; that generalization is not developed here.

The frame parametrization $U_i = \exp(\phi_{x_i}) \exp(Y_i)$ of Section 2.2 has positive determinant, while the passive law of Eq. (12) is defined for arbitrary invertible frames. A single real exponential is not a global chart of $\mathrm{GL}^+(K)$: for $K \geq 2$ the exponential map of $\mathfrak{gl}(K, \mathbb{R})$ is neither injective nor surjective onto $\mathrm{GL}^+(K)$, and a product of real exponentials need not have a real logarithm (Culver, 1966). These restrictions determine which frames the parametrization reaches; they do not alter the endpoint law of the links. With block-diagonal coefficients, as in the evaluated runs, the same statements hold blockwise in $\prod_b \mathrm{GL}^+(d_b)$.

The comparison link of Proposition 1 is built from the same frames and reference maps.

Proof [Proof of Proposition 1] (1) By the definition of the reference laws, $(S_Q)_* \widetilde{q}_i = (S_Q U_i^{-1})_* q_i$ and $(S_K)_* \widetilde{q}_j = (S_K U_j^{-1})_* q_j$. Pushing both laws forward by the invertible map $U_i S_Q^{-1}$ sends the first to q_i and the second to $(U_i S_Q^{-1} S_K U_j^{-1})_* q_j$, and a common invertible pushforward leaves the divergence unchanged (Lemma 10), which gives Eq. (21). Under $U_i \mapsto g_i U_i$ the link becomes $g_i \Theta_{ij}^{\mathrm{cmp}} g_j^{-1}$. The redundancy leaves $(AS_Q)^{-1} A S_K = S_Q^{-1} S_K$ unchanged, and the common reference change gives $U_i a (S_Q a)^{-1} S_K a (U_j a)^{-1} = \Theta_{ij}^{\mathrm{cmp}}$, with reference representative $a^{-1} S_Q^{-1} S_K a$. The equation $U_i S_Q^{-1} S_K U_j^{-1} = U_i e^{L^{\mathrm{cmp}}} U_j^{-1}$ holds exactly when $S_Q^{-1} S_K = e^{L^{\mathrm{cmp}}}$, which is the exponential case of Definition 2 with an edge-independent generator, and $S_Q = S_K$ gives $U_i U_j^{-1} = \Omega_{ij}$. (2) The law $M_* q_j$ is concentrated on the affine subspace $M\mu_j + \operatorname{im} M$, which has dimension $\operatorname{rank} M < K$ and is therefore a Lebesgue-null set, while q_i has an everywhere positive density. Hence q_i assigns this subspace zero mass and $M_* q_j$ assigns it full mass, so neither law is absolutely continuous with respect to the other, and the divergence is infinite in both directions. A head map with $s_h < K$ has rank at most s_h. ∎

By Culver (1966), the real logarithm in item 1 exists exactly when every Jordan block of a negative eigenvalue of $S_Q^{-1} S_K$ occurs an even number of times.

Table 6 collects the symbols of states, frames, and interfaces used in the main text.

Appendix B. Proofs for Section 3

B.1 Proof of Proposition 3

Item 1 follows by substituting $U_i' = g_i U_i$ into Eq. (37). Item 2 follows from $\exp(a^{-1} L a) = a^{-1} \exp(L) a$, which holds term by term in the exponential series, so that $(U_i a) \exp(a^{-1} L_{ij} a)(U_j a)^{-1} = \Theta_{ij}$. Item 3 is the identity $\det e^L = e^{\operatorname{tr} L}$ (Hall, 2015). The characterization in item 4 is Theorem 1 of Culver (1966), and membership in $\mathrm{GL}^+(K)$ follows from item 3. For the last statement, $\operatorname{diag}(-1, -2) \oplus I_{K-2}$ has positive determinant and two negative eigenvalues, each with a single Jordan block of size one, so it has no real logarithm.

When the reference representative has no eigenvalue on the closed negative real axis, it has a unique real logarithm whose eigenvalues lie in the strip $|\operatorname{Im} z| < \pi$, the principal logarithm (Higham, 2008, Thm. 1.31). The principal logarithm is therefore a chart on the open subset of $\mathrm{GL}^+(K)$ without eigenvalues on the closed negative real axis; it is not a global chart of the group. In lattice gauge theory the link e^{aA} has the algebra-valued potential A as its logarithmic coordinate, and A transforms as a connection under local

Symbol	Meaning, domain, and role
N, K, H, B, N_ℓ	Number of agents (positions), state dimension, heads per layer, frame blocks, and layers
q_i, p_i	Gaussian belief and prior of agent i, points of the statistical fiber $\mathcal{G}(E_i)$
ξ_i, C_i	Reference moments $U_i^{-1}\mu_i$ and $U_i^{-1}\Sigma_i U_i^{-\top}$
U_i	Frame $\mathsf{V} \to E_i$
$\pi_{ij}, \beta_{ij}^{(h)}, \tau_h$	Source prior, routing weight of head h, and positive headwise temperature
$\mathcal{N}_i$	Allowed senders of receiver i, the support of π_i; causally $\{0, \dots, i\}$
$S_{Q,h}, S_{K,h}$	Query map (receiver side) and key map (sender side), $\mathsf{V} \to \mathcal{Y}_h = \mathbb{R}^{r_h}$, full row rank
$A_{Q,i,h}, A_{K,i,h}$	Local measurements $S_{Q,h}U_i^{-1}, S_{K,h}U_i^{-1} : E_i \to \mathcal{Y}_h$
T_h, O_h	Sending map $\mathsf{V} \to \mathcal{Z}_h = \mathbb{R}^{s_h}$ (sender side) and stored receiving factor $\mathcal{Z}_h \to \mathsf{V}$
$\widetilde{O}_h$	Applied receiving map (receiver side): O_h in the affine chart, otherwise a block of Eq. (43)
M_h	Reference head transfer $\widetilde{O}_h T_h$
$\boldsymbol{\beta}_{ij}$	Head-diagonal routing matrix $\bigoplus_h \beta_{ij}^{(h)} I_{s_h}$ acting on the stacked content spaces
c_{res}	Residual coefficient in $\{0, 1\}$

Table 6: Symbols for states, frames, and interfaces. Comparison and content spaces are distinct even when their dimensions agree, and layer indices are suppressed. The edge objects are listed in Table 1.

gauge transformations (Creutz, 1983); in Definition 2 the frames dress the potential instead, which is why the generator is invariant under independent site changes of frame. The flat identification of Remark 4 is trivialized explicitly by the global reference space, and a finite discrete base has no overlap or continuum topology obstruction; the communication graph may nevertheless carry independent connection data with nonidentity cycle products, so the identity of that remark applies to Ω and not to every link.

B.2 The two orders of a single-head link

Contracting the scaled link $\Theta_{ij}(s) = U_i e^{sO_hT_h} U_j^{-1}$ of Proposition 6 against one routing row at fixed source states gives

$$\sum_j \beta_{ij}\Theta_{ij}(s)\mu_j = \sum_j \beta_{ij}\Omega_{ij}\mu_j + s\sum_j \beta_{ij}U_iO_hT_hU_j^{-1}\mu_j + \mathcal{O}(s^2). \tag{74}$$

The zeroth-order term transports complete source states by the flat identification. It is the message of complete-state alignment, the comparison with $S_Q = S_K = I$ in Proposition 1 (Section 6.1). The first-order term is the predictive message of Eq. (28) in the affine chart. Both message forms are thus orders of one exponential link. This is a statement about the form of the message and does not recover either objective. With an independent source label for each head, the zeroth-order term becomes a head-routed flat transport (Theorem 7).

B.3 Proof of Theorem 7

Throughout, the heads tile the reference space, $\bar{s} = K$, the stacked sending map $T \in \mathbb{R}^{K\times K}$ is invertible, and $L = OT$.

Item 1. The relations $\Delta_h\Delta_g = \delta_{hg}\Delta_h$ and $\sum_h \Delta_h = I$ survive conjugation by T, and Eq. (44) is $\boldsymbol{\beta}_{ij} = \sum_h \beta_{ij}^{(h)}\Delta_h$ conjugated by T. Writing T^{-1} by block columns gives $\Pi_h = (T^{-1})_{:,h}T_h$. Since $\Delta_h Tx$ has the single nonzero block $T_h x$, $\Pi_h x = 0$ exactly when $T_h x = 0$, and $x \in \operatorname{im}\Pi_h = T^{-1}\operatorname{im}\Delta_h$ exactly when $T_g x = 0$ for every $g \neq h$. Finally, $\Pi(\boldsymbol{\beta}_{ij})$ is similar through T to the diagonal matrix $\boldsymbol{\beta}_{ij}$, whose eigenvalue $\beta_{ij}^{(h)}$ has the eigenspace $\operatorname{im}\Delta_h$ of dimension s_h. Conjugation by T^{-1} carries this eigenspace to $\operatorname{im}\Pi_h$, and the determinant is the product of the eigenvalues.

Item 2. In the affine chart $\widetilde{O}T = OT = L$, and in the first-order chart $(T^{-1}+O)T = I + L$. In the exponential chart, Lemma 5 with $A = O$ and $B = T$ gives $e^L = I + O\varphi_1(TO)T$, and multiplying on the right by T^{-1} gives both forms of $\widetilde{O}$ in Eq. (45), together with $\widetilde{O}T = e^L$. Since $T(OT)^n = (TO)^n T$ for every n, $T\widetilde{O} = Te^{OT}T^{-1} = e^{TO}$, and $\det e^{TO} = e^{\operatorname{tr}(TO)} = e^{\operatorname{tr} L}$.

Item 3. Block multiplication gives $\sum_h \beta_{ij}^{(h)}\widetilde{O}_h T_h = \widetilde{O}\boldsymbol{\beta}_{ij}T$, the reference representative of the routed edge map of Eq. (32). Inserting TT^{-1} and using item 2,

$$\widetilde{O}\boldsymbol{\beta}_{ij}T = (\widetilde{O}T)(T^{-1}\boldsymbol{\beta}_{ij}T) = f(L)\Pi(\boldsymbol{\beta}_{ij}). \tag{75}$$

Conjugating by the frames and summing over the sources gives Eq. (46).

Item 4. With $f(L) = e^L$, $U_i e^L \Pi(\boldsymbol{\beta}_{ij})U_j^{-1} = (U_i e^L U_j^{-1})(U_j\Pi(\boldsymbol{\beta}_{ij})U_j^{-1})$, which is Eq. (47). The determinant of $\mathring{\mathcal{M}}_{ij} = e^L\Pi(\boldsymbol{\beta}_{ij})$ follows from item 1 and $\det e^L = e^{\operatorname{tr} L}$. If every $\beta_{ij}^{(h)}$ is positive, this determinant is positive, so $\mathring{\mathcal{M}}_{ij} \in \mathrm{GL}^+(K)$ and $\mathcal{M}_{ij} = U_i\mathring{\mathcal{M}}_{ij}U_j^{-1}$ is invertible. Shared routing has $\boldsymbol{\beta}_{ij} = \beta_{ij}I_K$, hence $\Pi(\boldsymbol{\beta}_{ij}) = \beta_{ij}I_K$, and Eq. (46) gives $\mathcal{M}_{ij} = \beta_{ij}U_i f(L)U_j^{-1}$. For the three choices of f this is $\beta_{ij}\Theta_{ij}$, $\beta_{ij}U_i L U_j^{-1}$, and $\beta_{ij}(\Omega_{ij} + U_i L U_j^{-1})$.

Item 5. Under $O \mapsto sO$ the generator becomes sL and the exponential-chart receiving map becomes $e^{sL}T^{-1}$. Since $L^nT^{-1} = (OT)^nT^{-1} = O(TO)^{n-1}$ for $n \geq 1$, the exponential series gives Eq. (49), which converges absolutely for every s because the exponential series does, and the first form of Eq. (49) is Eq. (45) for the factor sO. Expanding $\mathring{\mathcal{M}}_{ij}(s) = e^{sL}\Pi(\boldsymbol{\beta}_{ij})$ term by term at fixed routing, the order-zero term is $\Pi(\boldsymbol{\beta}_{ij})$, given by item 1. At order one $L\Pi(\boldsymbol{\beta}_{ij}) = OTT^{-1}\boldsymbol{\beta}_{ij}T = O\boldsymbol{\beta}_{ij}T$, and at order two $\frac{1}{2}L^2\Pi(\boldsymbol{\beta}_{ij}) = \frac{1}{2}OTOTT^{-1}\boldsymbol{\beta}_{ij}T = \frac{1}{2}O(TO)\boldsymbol{\beta}_{ij}T$. Because $\boldsymbol{\beta}_{ij}$ is head diagonal, the (g,h) block of $(TO)\boldsymbol{\beta}_{ij}$ is $(T_gO_h)\beta_{ij}^{(h)}$, which gives the head expansion of Eq. (50). Under shared routing the order-zero reference map is $\beta_{ij}I_K$, and $U_i\beta_{ij}\xi_j = \beta_{ij}\Omega_{ij}\mu_j$.

Remark 16 (No complement) *Under the hypotheses of Theorem 7, if O is also invertible, the complement projector $O(TO)^{-1}T$ of Lemma S2 equals I_K and $Oe^{TO}(TO)^{-1} = e^LT^{-1}$, so the complement-preserving aggregation of Proposition S3 has no complement term, and its message is the exponential-chart message of item 3 of Theorem 7.*

Proof With O and T invertible, $O(TO)^{-1}T = OO^{-1}T^{-1}T = I$, and $O(TO)^nO^{-1} = (OT)^n$ gives $Oe^{TO}(TO)^{-1} = Oe^{TO}O^{-1}T^{-1} = e^{OT}T^{-1}$. The complement term of Eq. (S5) therefore vanishes, and its head-sum term, with receiving blocks taken from $Oe^{TO}(TO)^{-1} = e^LT^{-1}$,

is the exponential-chart message of item 3. ■

The same message is the mean of a head-labeled transport with one pinned generator. If each head draws its own source label J^h with probability $\beta_{ij}^{(h)}$, independently across heads, and the reference message is $\sum_h e^L \Pi_h \xi_{J^h}$, then by Eq. (44) its mean is $\sum_h \sum_j \beta_{ij}^{(h)} e^L \Pi_h \xi_j = e^L \sum_j T^{-1} \boldsymbol{\beta}_{ij} T \xi_j = \sum_j \widetilde{O} \boldsymbol{\beta}_{ij} T \xi_j$. This is the static limit of the latent-link model of Supplement S6 (Remark S17). Only the edge maps are group valued: the aggregate over sources is a predictive mean, a combination of transported vectors with routing weights (Theorem 15), and not a group operation. In the affine chart the transport is the frame-dressed Lie-algebra element $U_i L U_j^{-1}$, a matrix that may be singular, and in the first-order chart it is $U_i(I+L)U_j^{-1} = \Omega_{ij} + U_i L U_j^{-1}$.

Residuals. The complement-preserving aggregation of Proposition S3 keeps the receiver's state on the directions that no head writes, and by Remark 16 that complement is empty when the heads tile, so the receiver's own state enters only through the self-weights $\beta_{ii}^{(h)}$. A projected residual requires channels that do not tile, $\bar{s} < K$, whose unread coordinates pass through the layer unchanged, the role the residual stream plays for subspaces that no head reads or writes (Elhage et al., 2021). The additive residual $c_{\mathrm{res}}\mu_i$ of Eq. (28) is the residual connection of residual networks (He et al., 2016), an architectural term that this aggregation does not produce.

Reach of the charts. The orders do not enlarge the family of mean maps. For invertible T the map $O \mapsto T^{-1} + O$ is a bijection, so the first-order truncation reaches every receiving map, as the affine chart does. Since OT ranges over all of $\mathfrak{gl}(K)$, the exponential chart reaches a receiving map $\widetilde{O}$ exactly when $\widetilde{O}T$ has a real logarithm L, with $O = LT^{-1}$, and by Proposition 3 this forces $\det(\widetilde{O}T) > 0$. The three parametrizations therefore represent the same tiled shared-generator mean maps up to the real-logarithm domain. They differ in what the zero of the stored factor means (zero content in the affine chart; in the first-order chart, which has the affine function class, a decay of the stored factor pulls toward the head-routed flat transport rather than toward zero content), in the reachable component, and in the conditioning and optimization geometry of the trained links, and these are the properties that a comparison between them tests.

The identities were also evaluated in float64 by the script `check_routed_transport.py`, distributed with the manuscript source, at $K = 6$ with three heads of width two and four causal positions, a random invertible T, random O and frames, and random positive routing weights. Items 1 to 5 and Remark 16 hold to a relative error of at most 2×10^{-15}, apart from the determinant identities, which hold to 1.3×10^{-14}; the factorization of Eq. (47) holds to 8×10^{-16}, the computed eigenvalues of $\Pi(\boldsymbol{\beta}_{ij})$ agree with the head weights to 4×10^{-16}, and every reference representative $\mathring{\mathcal{M}}_{ij}$ had positive determinant. These sizes test the formulas and carry no information about trained models.

B.4 Two charts of one group

The phrase "first-order channel" could suggest that the affine head map is a weak-field truncation of the exponential link. On a specified logarithm domain the relation is instead an exact change of coordinates. In this appendix the width $s \leq K$ is arbitrary. Fix

$O \in \mathbb{R}^{K\times s}$ and $T \in \mathbb{R}^{s\times K}$ with TO invertible, so that $\mathbb{R}^K = W \oplus W'$ with $W = \operatorname{im} O$ and $W' = \ker T$, and let $\Pi = O(TO)^{-1}T$ be the projector onto W along W'. For $B \in \mathrm{GL}(s)$ put

$$\Upsilon(B) = (I - \Pi) + OB(TO)^{-1}T, \tag{76}$$

the operator that fixes W' pointwise and acts on W as B in the basis formed by the columns of O.

Proposition 17 (Two charts of one group) *1. $\Upsilon : \mathrm{GL}(s) \to \mathrm{GL}(K)$ is an injective homomorphism with $\det \Upsilon(B) = \det B$. Its image is the factor of the block-diagonal Levi subgroup $\mathrm{GL}(W) \times \mathrm{GL}(W')$ that acts trivially on W'.*

2. *Linear links: for $C \in \mathbb{R}^{s\times s}$, $I + OCT = \Upsilon(I_s + CTO)$, so $\det(I + OCT) = \det(I_s + CTO)$.*

3. *Exponential links: $\exp(OT) = \Upsilon(e^{TO})$.*

4. *Exact conversion: let $O_{\mathrm{lin}} \in \mathbb{R}^{K\times s}$ be such that $I_s + TO_{\mathrm{lin}}$ has no eigenvalue on $(-\infty, 0]$, put $Z = \log(I_s + TO_{\mathrm{lin}})$ with the principal logarithm, and put $O = O_{\mathrm{lin}}\varphi_1(Z)^{-1}$. Then $\varphi_1(Z)$ is invertible, $TO = Z$, and*

$$\exp(OT) = I_K + O_{\mathrm{lin}}T. \tag{77}$$

Item 4 says that the flat identification plus a linear channel term, $\Omega_{ij} + U_i O_{\mathrm{lin}} T U_j^{-1}$, is exactly an exponential group link for every O_{lin} in an open set containing $O_{\mathrm{lin}} = 0$. On their common domain the linear and logarithmic parametrizations describe the same group elements, related by $I_s + TO_{\mathrm{lin}} = e^{TO}$, and Proposition 6 states that the two charts agree to first order at the identity. Using one numerical generator in both $I + L$ and e^L is a first-order approximation only: at $L = I$, $e^L = eI$ while $I + L = 2I$. The charts differ in three further ways. The linear parameter space contains the singular matrices, and a step in (O_{lin}, T) can carry $\det(I_K + O_{\mathrm{lin}}T) = \det(I_s + TO_{\mathrm{lin}})$ through zero, whereas the logarithmic chart cannot reach that boundary and, by Proposition 3, also excludes the matrices without a real logarithm, including every matrix of negative determinant. An optimizer acting on the factors induces different changes of the composite matrix in each chart, so the two charts carry different optimization geometries. Finally, Proposition 17 concerns an unrouted operator with one generator; routed head sums are covered by Theorem 7, and edge-dependent generators leave the one-generator family. When the heads tile the state, O_{lin} is the stored factor of the first-order chart, whose effective receiving map is $T^{-1} + O_{\mathrm{lin}}$.

Proof of Proposition 17. For item 1, $\Upsilon(B)Oy = OBy$ because $(I - \Pi)O = 0$ and $(TO)^{-1}TO = I$, and $\Upsilon(B)w' = w'$ for $w' \in W'$ because $\Pi w' = 0$. An operator on $W \oplus W'$ is determined by its action on the two summands, so $\Upsilon(B)\Upsilon(B') = \Upsilon(BB')$, $\det \Upsilon(B) = \det B$, and $\Upsilon(B) = I$ only for $B = I$. Conversely, an invertible operator that maps W to itself and fixes W' pointwise acts on W through some $B \in \mathrm{GL}(s)$ in the basis formed by the columns of O, and therefore equals $\Upsilon(B)$; this identifies the image. For item 2, $(I + OCT)Oy = O(I + CTO)y$ and $(I + OCT)w' = w'$. For item 3, $(OT)^n O = O(TO)^n$ gives $\exp(OT)Oy = Oe^{TO}y$, and $\exp(OT)w' = w'$. For item 4, the eigenvalues of Z have imaginary parts in $(-\pi, \pi)$, while the eigenvalues $(e^z - 1)/z$ of $\varphi_1(Z)$ vanish only at $z \in$

$2\pi i\mathbb{Z} \setminus \{0\}$, so $\varphi_1(Z)$ is invertible. Since $TO_{\mathrm{lin}} = e^Z - I = Z\varphi_1(Z)$, with Z and $\varphi_1(Z)$ commuting, $TO = TO_{\mathrm{lin}}\varphi_1(Z)^{-1} = Z$. Lemma 5 then gives $\exp(OT) = I + O\varphi_1(TO)T = I + O_{\mathrm{lin}}\varphi_1(Z)^{-1}\varphi_1(Z)T = I + O_{\mathrm{lin}}T$.

B.5 The standard layer at fixed routing

Three qualifications fix the scope of the correspondence of Section 3.4. First, $\exp(\mathfrak{L}_\beta)$ composes routed messages along multi-step paths within one layer, so it differs from the per-edge link with routing outside the exponent of Theorem 7 (shown below). Second, at a single edge the standard layer $\delta_{ij}I + O\boldsymbol{\beta}_{ij}T$ is neither the tiled exponential message nor that message's first-order truncation, whose zeroth order is the head-routed flat transport rather than the identity on the receiver; the routing-independent receiver term ξ_i equals $\sum_j \widetilde{O}\boldsymbol{\beta}_{ij}T\xi_j$ for all states only when every head routes the receiver to itself with unit weight (shown below). Third, the routing weights depend on the states, so the layer is linear only at fixed routing. Outside the principal-logarithm domain the layer is not the exponential of a principal real generator.

Whether W_OW_V is invertible does not decide the group structure. For $H \geq 2$ a single head has $\operatorname{rank} W_O^{(h)}W_V^{(h)} \leq K/H < K$ and is singular, while the stacked W_OW_V is invertible for generic weights but is not constrained to be invertible by training. The exponential chart $\widetilde{O} = e^{OT}T^{-1}$ is defined on the set of invertible T. Combining the exponential chart with $c_{\mathrm{res}} = 1$, as the language model of Section 7.2 does, adds the receiver twice, once through the self-weights $\beta_{ii}^{(h)}$ inside the zeroth order and once through the residual; this paper gives no derivation of that combination.

At fixed routing weights the residual multi-head layer maps ξ_i to $\xi_i + \sum_j \sum_h \beta_{ij}^{(h)} M_h \xi_j$, which is the ith block of $(I_{NK} + \mathfrak{L}_\beta)\boldsymbol{\xi}$ with $\mathfrak{L}_\beta = \sum_h B^{(h)} \otimes M_h$; the (i,j) block of $\mathfrak{L}_\beta$ is $\sum_h \beta_{ij}^{(h)} M_h = O\boldsymbol{\beta}_{ij}T$. With frames the blocks are $U_i(\sum_h \beta_{ij}^{(h)} M_h)U_j^{-1} = \mathcal{M}_{ij}$, so $g\mathfrak{L}_\beta g^{-1}$ has the blocks $g_i\mathcal{M}_{ij}g_j^{-1}$ for $g = \bigoplus_i g_i$. Causal routing has $\beta_{ij}^{(h)} = 0$ for $j > i$, so $\mathfrak{L}_\beta$ is block lower triangular, $I + \mathfrak{L}_\beta$ is invertible exactly when every self-block $I_K + \sum_h \beta_{ii}^{(h)} M_h$ is, and the determinant of $I + \mathfrak{L}_\beta$ is the product of the determinants of its diagonal blocks $I_K + \sum_h \beta_{ii}^{(h)} M_h$ (conjugated by U_i when frames are present). The first position has itself as its only source, so $\beta_{00}^{(h)} = 1$ for every head and its self-block is $I_K + \sum_h M_h = I_K + OT$.

The spectrum of $I + \mathfrak{L}_\beta$ is the union of the spectra of its diagonal blocks. If none of these meets $(-\infty, 0]$, the principal logarithm of $I + \mathfrak{L}_\beta$ exists, is real, and satisfies $\exp(\log(I + \mathfrak{L}_\beta)) = I + \mathfrak{L}_\beta$ (Higham, 2008, Thm. 1.31). As a primary matrix function it equals $p(I + \mathfrak{L}_\beta)$ for a polynomial p that interpolates the logarithm on the spectrum (Higham, 2008, Ch. 1), and every polynomial in a block lower triangular matrix is block lower triangular. The square $(\mathfrak{L}_\beta^2)_{ik} = \sum_j (\mathfrak{L}_\beta)_{ij}(\mathfrak{L}_\beta)_{jk}$ receives a contribution from every intermediate position j with $k \leq j \leq i$, so $\exp(\mathfrak{L}_\beta)$ contains multi-step routed paths that the per-edge maps of Theorem 7 do not.

For the receiver term, suppose that $\sum_j \widetilde{O}\boldsymbol{\beta}_{ij}T\xi_j = \xi_i$ for all states. Then $\widetilde{O}\boldsymbol{\beta}_{ii}T = I_K$ and $\widetilde{O}\boldsymbol{\beta}_{ij}T = 0$ for $j \neq i$. The first equation makes $\widetilde{O}\boldsymbol{\beta}_{ii}$ invertible, so $\widetilde{O}$ is invertible and every $\beta_{ii}^{(h)}$ is positive; the second then gives $\boldsymbol{\beta}_{ij} = 0$ for $j \neq i$, so every head routes only to the receiver, $\boldsymbol{\beta}_{ii} = I_K$, and $\widetilde{O} = T^{-1}$. A float64 evaluation at $K = 6$, $H = 3$,

and $N = 4$ with random frames, maps, and causal routing reproduces the block-triangular determinant identity to a relative error of 7×10^{-15}, the affine-chart message with $c_{\mathrm{res}} = 1$ to 4×10^{-16}, and a principal logarithm whose strictly upper blocks are below 2×10^{-15} (script of Appendix B.3).

B.6 Proof of Proposition 9

For item 1 with $i - j \geq 2$, flatness with middle index $i - 1$ gives $\Theta_{ij} = \Theta_{i,i-1}\Theta_{i-1,j}$, and induction on $i - j$ gives the ordered product; conversely, ordered products of adjacent matrices are associative. For item 2, set $g_0 = I$ and $g_i = \Theta_{i0}$. By item 1, $g_i g_j^{-1} = \Theta_{i,i-1} \cdots \Theta_{j+1,j} = \Theta_{ij}$, and the converse is the cocycle identity $g_i g_j^{-1} g_j g_k^{-1} = g_i g_k^{-1}$. The projector satisfies $PP = P$ but is singular. The triangle holonomies of Definition 8 thus detect failure of the cocycle condition on the complete ordered graph, but their separate traces or spectra need not form a complete set of invariants. Inverse-edge completion supplies these loop observables without contradicting the absence of directed cycles on causal support.

B.7 Chart initialization and evaluation of φ_1

At the block-projector initialization $T = I$, the content $c_{\mathrm{out}}\boldsymbol{\beta}_{ij}$ at output scale $c_{\mathrm{out}} > 0$ requires $O = c_{\mathrm{out}} I$, $(c_{\mathrm{out}} - 1)I$, and $(\log c_{\mathrm{out}})I$ in the affine, first-order, and exponential charts, since $z\varphi_1(z) = e^z - 1$ equals $c_{\mathrm{out}} - 1$ at $z = \log c_{\mathrm{out}}$. At $c_{\mathrm{out}} = 1$ both centered charts start at $O = 0$, whereas reusing $O = I$ in the exponential chart would start the content at $e\boldsymbol{\beta}_{ij}$. The factor $\varphi_1(TO)$ needs no inverse of TO: for every square A,

$$\exp \begin{pmatrix} A & I \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} e^A & \varphi_1(A) \\ 0 & I \end{pmatrix}, \tag{78}$$

because $\left(\begin{smallmatrix} A & I \\ 0 & 0 \end{smallmatrix}\right)^n = \left(\begin{smallmatrix} A^n & A^{n-1} \\ 0 & 0 \end{smallmatrix}\right)$ for $n \geq 1$. This is the two-block case of the block-triangular exponential formula of Van Loan (1978), so one exponential of size $2\bar{s}$ returns e^{TO} and $\varphi_1(TO)$.

Appendix C. Proofs for Sections 4 to 6

C.1 Proof of Theorem 11

The four items are proved in order; the notation is that of Section 4.

Proof (1) The comparison laws of Eqs. (17) and (18) are built from $S_{Q,h}\xi_i$, $S_{Q,h}C_iS_{Q,h}^\top$, and the corresponding key moments, so the costs and the Gibbs weights of Eq. (24) depend on the states only through the reference moments. Applying U_i^{-1} to Eq. (28) gives the stated reference form. (2) Substituting the transformed variables into Eq. (11) gives $\xi_i' = \xi_i$ and $C_i' = C_i$, so by (1) the weights are unchanged. The head maps obey Eq. (30), which leaves only the external factors g_i and g_j^{-1}; the routed edge map combines head maps with invariant weights, and the message transforms by g_i. (3) Under the common reference change, $\xi_i \mapsto a^{-1}\xi_i$ and $C_i \mapsto a^{-1}C_i a^{-\top}$, so every projected mean $S_{Q,h}\xi_i$, every projected covariance $S_{Q,h}C_iS_{Q,h}^\top$, and the corresponding key moments are unchanged. In each chart the effective receiving map transforms as $\widetilde{O} \mapsto a^{-1}\widetilde{O}$: trivially

in the affine chart, as $(Ta)^{-1} + a^{-1}O = a^{-1}(T^{-1} + O)$ in the first-order chart, and as $e^{a^{-1}OTa}(Ta)^{-1} = a^{-1}e^{OT}T^{-1}$ in the exponential chart. Hence $M_h \mapsto a^{-1}M_h a$, and $U_i M_h \xi_j$ is unchanged. The local coordinates $\mu_i = U_i \xi_i$ are unchanged, so the operations applied in local coordinates are unaffected. Under A, both comparison laws are pushed forward by one invertible map, so Lemma 10 applied in $\mathcal{Y}_h$ leaves every $D_{ij}^{(h)}$ unchanged. Under the headwise change, the stacked factors become (BT, OB^{-1}) with $B = \bigoplus_h B_h$, so OT is unchanged, $(BT)^{-1} + OB^{-1} = (T^{-1} + O)B^{-1}$, and $e^{OT}(BT)^{-1} = e^{OT}T^{-1}B^{-1}$; in each chart $\widetilde{O}_h \mapsto \widetilde{O}_h B_h^{-1}$, so every M_h is unchanged, and the messages depend on the content maps only through the M_h. (4) Intermediate factors $g_{i_k}^{-1} g_{i_k}$ cancel in the loop product, and cyclicity of the trace makes its trace invariant. Expanding $U_{i_0}^{-1}\mathcal{H}_\gamma U_{i_0} = \mathring{\mathcal{M}}_{i_0 i_1} \cdots \mathring{\mathcal{M}}_{i_{n-1} i_0}$ with the reference representatives $\mathring{\mathcal{M}}_{ij} = U_i^{-1}\mathcal{M}_{ij}U_j = \sum_h \beta_{ij}^{(h)} M_h$ gives the polynomial. Each factor $\mathcal{M}_{i_k i_{k+1}}$ requires $i_{k+1} \le i_k$ on the causal support, so the indices are nonincreasing along the walk and a closed walk is constant. ■

C.2 Interface redundancies and balance laws

Item (3) of Theorem 11 has the following consequences. When $S_{Q,h} = S_{K,h}$, the shared comparison map enters only through its row space. The cogredient redundancy $(S_{Q,h}, S_{K,h}) \mapsto (AS_{Q,h}, AS_{K,h})$ differs from the contragredient redundancy $(S_{Q,h}, S_{K,h}) \mapsto (AS_{Q,h}, A^{-\top}S_{K,h})$ of the dot-product score $\xi_i^\top S_{Q,h}^\top S_{K,h} \xi_j$; the two share only $A \in \mathrm{O}(r_h)$, the residual freedom of whitening in Proposition 14. In particular, a common rescaling of $S_{Q,h}$ and $S_{K,h}$ does not change the routing temperature. Only $\mathcal{Y}_h$ and $\mathcal{Z}_h$ as abstract vector spaces are determined by the model; their bases are internal parameter redundancies. Gradient flow on a loss $\mathcal{E}$ that depends on these maps only through the costs and messages therefore conserves $S_{Q,h}S_{Q,h}^\top + S_{K,h}S_{K,h}^\top$ and, in each chart, $T_h T_h^\top - O_h^\top O_h$ (Du et al., 2018; Kunin et al., 2021). For the first law, invariance of $\mathcal{E}$ under $(S_{Q,h}, S_{K,h}) \mapsto (e^{\epsilon Z}S_{Q,h}, e^{\epsilon Z}S_{K,h})$ for every $Z \in \mathfrak{gl}(r_h)$ gives $G := (\nabla_{S_{Q,h}}\mathcal{E})S_{Q,h}^\top + (\nabla_{S_{K,h}}\mathcal{E})S_{K,h}^\top = 0$, and along $\dot{S} = -\nabla_S \mathcal{E}$ the time derivative of $S_{Q,h}S_{Q,h}^\top + S_{K,h}S_{K,h}^\top$ is $-(G + G^\top) = 0$. For the second, invariance under $(T_h, O_h) \mapsto (e^{\epsilon Z}T_h, O_h e^{-\epsilon Z})$ for every $Z \in \mathfrak{gl}(s_h)$ gives $G' := (\nabla_{T_h}\mathcal{E})T_h^\top - O_h^\top(\nabla_{O_h}\mathcal{E}) = 0$, and the time derivative of $T_h T_h^\top - O_h^\top O_h$ is $-(G' + G'^\top) = 0$. Adaptive optimizers without weight decay can drift along these directions. The second law requires invariance of the entire loss under $\mathrm{GL}(s_h)$; the channel floor of the receiver of Supplement S5 reduces that invariance to the orthogonal group.

C.3 Proofs of Lemma 10 and Corollary 12

Proof [Proof of Lemma 10] Both densities acquire the same Jacobian factor, which cancels in their ratio (Amari, 2016). The same argument applies to f-divergences, and the data-processing inequality applied to A and A^{-1} gives equality for every Rényi order (van Erven and Harremøes, 2014, Thm. 9). ■

Proof [Proof of Corollary 12] Apply U_i^{-1} to Eqs. (34) and (36), and use item (1) of Theorem 11 for the attention step. The frames are depth independent, so passthrough keeps $C_i^{(0)}$, and Theorem 15 gives the moment-matched covariance in reference coordinates. ■

Of the four frame channels named after Corollary 12, the conjugated normalization and MLP and the frame-dependent decoder have covariant alternatives: conjugating the block, $\mu_i \mapsto U_i G(U_i^{-1}\mu_i)$, or scaling each frame block by a function of the invariant $\mu_b^\top \Sigma_{bb}^{-1}\mu_b$, gives a block covariant under the passive site action, and with such a block and a decoder acting on ξ_i the predictor depends on the frames only through $\xi_i^{(0)}$ and C_i.

C.4 Proof of Proposition 14

Proof Set $P_j = V_*^{-1}$ in Eq. (59). The trace, query quadratic, dimension, and determinant terms are independent of j, except for $b_j^\top V_*^{-1} b_j$ and the cross term. Thus

$$D_{ij} = c_i + \tfrac{1}{2}\|\mathsf{k}_j\|^2 - \mathsf{q}_i^\top \mathsf{k}_j, \tag{79}$$

for a row constant c_i. Substitution into the categorical solution $\mathrm{softmax}_j(-D_{ij}/\tau + \log \pi_{ij})$ removes c_i. Under the stronger isotropic assumption the trace and log-determinant terms cancel and the remaining quadratic is the stated squared distance. ■

C.5 Proofs for Section 6

Variational source selection. For the laws of Eqs. (64) and (65), the log density ratio on the event $J = j$ is $\log(\beta_{ij}/\pi_{ij}) + \log\big(q_{Q,i}(y)/p_{K,j}(y)\big)$. Integrating over $y \sim q_{Q,i}$ and summing over j with weights β_{ij} gives Eq. (66). At a declared temperature $\tau > 0$, use

$$\mathcal{A}_i(\beta) = \sum_{j \in \mathcal{N}_i} \beta_{ij} D_{ij} + \tau D_{\mathrm{KL}}(\beta_i \| \pi_i). \tag{80}$$

The Hessian $\tau\,\mathrm{diag}(1/\beta_{ij})$ is positive definite on nonzero simplex tangent directions. The Lagrange condition is $D_{ij} + \tau(\log(\beta_{ij}/\pi_{ij}) + 1) + \lambda_i = 0$. For finite costs the unique minimizer is interior on $\mathcal{N}_i$, because the derivative of the entropy term diverges at the boundary of the simplex, and it is Eq. (68); for uniform π_i,

$$\beta_{ij}^* = \mathrm{softmax}_{j \in \mathcal{N}_i}(-D_{ij}/\tau). \tag{81}$$

Substitution gives the minimum $-\tau \log Z_i$, with $Z_i = \sum_{j \in \mathcal{N}_i} \pi_{ij} \exp(-D_{ij}/\tau)$, by the usual entropy-regularized variational calculation (Boyd and Vandenberghe, 2004; Wainwright and Jordan, 2008; Niculae and Blondel, 2017). The categorical minimizer needs no gradient-based update of q_i.

Comparison through the content link. In the comparison of Proposition 1, complete-state alignment $S_Q = S_K = I_K$ gives

$$D_{ij} = D_{\mathrm{KL}}\big(q_i \| (\Omega_{ij})_* q_j\big), \tag{82}$$

and square invertible comparison maps replace Ω_{ij} by $\Theta^{\mathrm{cmp}}_{ij} = U_i S_Q^{-1} S_K U_j^{-1}$, whose generator, when it exists, is shared by every edge because the comparison maps are shared by the layer; edge-dependent comparison generators would require edge-dependent comparison maps. This comparison link equals the content link $\Theta_{ij} = U_i e^L U_j^{-1}$ of Definition 2, with the shared generator $L = OT$, exactly when $S_Q^{-1} S_K = e^L$, that is, when $S_K = S_Q e^L$, and then

$$D_{ij} = D_{\mathrm{KL}}\big(q_i \| (\Theta_{ij})_* q_j\big), \tag{83}$$

so the receiver compares against the same transport that carries content; Eq. (82) is the pure-gauge case $L^{\mathrm{cmp}} = 0$. For $r_h < K$ no comparison link exists, by item 2 of Proposition 1, which is why the receiver is measured in the comparison space by S_Q. The data-processing inequality nevertheless bounds such a comparison above by Eq. (83) when $S_K = S_Q e^L$ and by Eq. (82) when $S_K = S_Q$ (Supplement S4). The evaluated model has $r_h < K$, so it forms no comparison link, and it learns $S_{Q,h}$, $S_{K,h}$, and the content maps (T_h, O_h) independently, so it does not impose Eq. (83).

Proof [Proof of Theorem 15, item 2] Write $X = c_{\mathrm{res}} k_i + \sum_h M^{(h)}_{iJ^{(h)}} k_{J^{(h)}}$. Conditioning on the labels gives $\mathbb{E}[X \mid J] = c_{\mathrm{res}} \mu_i + \sum_h M^{(h)}_{iJ^{(h)}} \mu_{J^{(h)}}$, whose average over the independent labels is μ_i^{att}. By the law of total covariance, $\operatorname{Cov} X = \mathbb{E} \operatorname{Cov}(X \mid J) + \operatorname{Cov} \mathbb{E}(X \mid J)$. Given the labels, each distinct source s contributes $A_s \Sigma_s A_s^\top$, with $A_i = c_{\mathrm{res}} I + \sum_h \mathbf{1}[J^{(h)} = i] M^{(h)}_{ii}$ and $A_s = \sum_h \mathbf{1}[J^{(h)} = s] M^{(h)}_{is}$ for $s \neq i$. Averaging the indicator products with $\mathbb{E}\big(\mathbf{1}[J^{(h)} = s]\mathbf{1}[J^{(h')} = s]\big) = c^{hh'}_{is}$, and using $c^2_{\mathrm{res}} = c_{\mathrm{res}}$, gives the terms in C_i, $\beta^{(h)}_{ii}$, and $c^{hh'}_{ij}$, and independence of the labels across heads gives the spread term. Conjugating by U_i^{-1} removes every frame, since $M^{(h)}_{ij} \Sigma_j M^{(h')\top}_{ij} = U_i M_h C_j M_{h'}^\top U_i^\top$, and adding Q_i completes the kernel covariance. ∎

Covariance contracts and the floor. Covariance passthrough with $c_{\mathrm{res}} = 1$ retains only the term C_i of Eq. (70). The implemented moment-matched contract evaluates Eq. (70) with the reference floor $\widetilde{Q}_i = \epsilon_Q \operatorname{Tr}(C_i) I/K$, so that $Q_i = \epsilon_Q \operatorname{Tr}(C_i) U_i U_i^\top / K$ transforms as a covariance because $\operatorname{Tr} C_i$ is invariant under the passive action. The output then satisfies $\lambda_{\min}(C_i^{\mathrm{att}}) \geq \epsilon_Q \operatorname{Tr}(C_i)/K$ whatever the routing and the rank of M_h, because the remaining terms of Eq. (70) together equal $\operatorname{Cov}(U_i^{-1} X)$ in the proof above and are therefore positive semidefinite. With $c_{\mathrm{res}} = 0$ the covariance is that of the message alone plus the floor; with $c_{\mathrm{res}} = 1$ the self-edge cross terms $\beta^{(h)}_{ii}(M_h C_i + C_i M_h^\top)$ can be indefinite, so no monotonicity in depth holds in general. When only the local diagonal of the covariance is stored, matching marginal variances is the projection that minimizes $D_{\mathrm{KL}}(p\|q)$ over diagonal Gaussian q, and it does not commute with arbitrary $\mathrm{GL}(K)$ congruences, so the passive law applies before that projection.

Appendix D. Positional and Masking Priors

With a nonuniform prior, Eq. (68) adds $\log \pi_{ij}$ to the logits with unit coefficient while the cost is divided by τ, so temperature never rescales a positional preference. Masks and positional biases therefore enter as priors over sources. This identification, and the entropy-

regularized or transport form of the softmax that underlies it, is established in the attention literature (Niculae and Blondel, 2017; Litman, 2025; Bianchessi et al., 2026; Litman and Guo, 2026). On the causal support $\mathcal{N}_i = \{0, \dots, i\}$, the uniform prior $\pi_{ij} = \mathbf{1}[j \le i]/(i+1)$ gives the support and row normalization of GPT-family attention (Radford et al., 2019). A window of radius w, $\pi_{ij} \propto \mathbf{1}[i - w \le j \le i]$, gives the sliding-window support of Longformer and Mistral (Beltagy et al., 2020; Jiang et al., 2023). The prior $\pi_{ij} \propto \exp[-\alpha_h(i-j)]$ gives the linear biases of attention with linear biases (ALiBi) (Press et al., 2022), and $\log \pi_{ij} = f_h(i-j) + \text{const}$ gives learned relative biases, parameterized by sinusoids in the global positional bias of Transformer-XL and shared within logarithmic offset buckets in T5 (Dai et al., 2019; Raffel et al., 2020). Press et al. (2022) state the geometric slope schedule $\alpha_h = 2^{-8h/H}$; their reference implementation uses it when H is a power of two and otherwise appends alternate slopes of the next power-of-two schedule. The construction fixes only the form in which a prior enters. It does not select a prior, except that the autoregressive factorization requires $\pi_{ik} = 0$ for $k > i$ at every layer, because each state depends on its own token.

Appendix E. The Linear Decoder as a Gibbs Readout

Let q_i be the Gaussian state after the chosen forward computation and p_v a normalized class template on the same latent space, with finite $D_{\mathrm{KL}}(q_i \| p_v)$. No convergence of q_i is assumed. Specify a categorical readout $\hat{p}_i$ by minimizing

$$\mathcal{J}_{\mathrm{decode}}(\hat{p}_i) = \sum_v \hat{p}_i(v) D_{\mathrm{KL}}(q_i \| p_v) + \tau_{\mathrm{dec}} D_{\mathrm{KL}}(\hat{p}_i \| \omega), \qquad \tau_{\mathrm{dec}} > 0, \tag{84}$$

where ω is a class prior. As for the routing row of Eq. (68), the unique optimum on its support is

$$\hat{p}_i(v) \propto \omega_v \exp\left[-D_{\mathrm{KL}}(q_i \| p_v)/\tau_{\mathrm{dec}}\right]. \tag{85}$$

This is a specified Gibbs readout. At unit temperature it is also the categorical mean-field coordinate update for a fixed q_i under the normalized mixture $p(v, x) = \omega_v p_v(x)$, as in the general expected-log update of Blei et al. (2017, §2.4, Eqs. 17–18).

The linear decoder of Eq. (36) is itself the unit-temperature Gibbs readout for the shared-covariance templates $p_v = \mathcal{N}(\Sigma_* w_v, \Sigma_*)$, with w_v the vth row of W_{out} and class prior $\log \omega_v = b_{\mathrm{out},v} + \kappa_{\mathrm{uni}} \log p_{\mathrm{uni}}(v) + \frac{1}{2} w_v^\top \Sigma_* w_v$ up to a constant, for any $\Sigma_* \succ 0$. Indeed, for $q_i = \mathcal{N}(\mu_i^{(N_\ell)}, \Sigma_i)$,

$$-D_{\mathrm{KL}}(q_i \| p_v) = w_v^\top \mu_i^{(N_\ell)} - \tfrac{1}{2} w_v^\top \Sigma_* w_v + c_i, \tag{86}$$

where c_i collects the terms $-\frac{1}{2}[\mathrm{Tr}(\Sigma_*^{-1}\Sigma_i) + \mu_i^{(N_\ell)\top} \Sigma_*^{-1} \mu_i^{(N_\ell)} - K + \log\det\Sigma_* - \log\det\Sigma_i]$, which do not depend on v. Adding $\log \omega_v$ gives the logits of Eq. (36) up to the class-independent c_i, so the belief covariance cancels from every logit difference. The templates are declared in the local coordinates in which the decoder acts; in reference coordinates the same readout is the frame-dependent decoder $W_{\mathrm{out}} U_i$ of Corollary 12, whereas templates declared in reference coordinates and compared with $(U_i^{-1})_* q_i$ would give a readout invariant under the passive action. Cross-entropy training of this readout does not become optimization of the full belief free energy merely because its scores are divergences. Supplement S8.3

develops the token-bank readout, which gives each class its own template covariance, and contrasts the Gibbs readout with the posterior-predictive average of a classifier.

Appendix F. Experimental Methods and Record

This appendix records the methods of the evaluated run of Section 7.2 beyond those stated in the main text, its parameter breakdown, numerical policies, diagnostics, and provenance, the configuration of the chart runs of Section 7.5, and the tables reserved for the final matched comparison. The records of the affine-chart runs at $K = 160$ and $K = 256$ are in Supplement S10, and the numerical identity checks of the theory are in Supplement S9.

F.1 Configuration

The run is configured by the selectors `attention_mode=projected_predictive`, which selects Gaussian-KL projected routing with full input and output scope followed by one predictive-content readout per layer, `relational_routing=gaussian_kl`, `content_link_chart=exponential`, `predictive_covariance=moment_matched` with floor 10^{-4}, `attention_residual=true`, `beta_attention_prior=causal`, `pos_phi=learned` with `pos_rotation=none`, `gauge_group=block_glk`, `decode_mode=linear` with a unigram prior of weight one, and `block_coordinates=local` and `decode_coordinates=local`, under which the normalization, the MLP, and the decoder act in local coordinates. The saved configuration lists separately the settings that the resolved path does not read, among them the receiver-law settings, the inner-step learning rates and trust regions, the self-coupling settings, and the settings of the prior-bank decoders; none of them affects the computation. The saved resolved-operator record confirms Gaussian-KL routing with exact full reduced covariances, the exponential content chart, the moment-matched covariance with a covariant floor, and routing on the pre-update states.

Diagonal storage is a partial gauge fixing: within each frame block every positive-definite covariance is congruent to a diagonal one and the frame absorbs the congruence, so with block-diagonal frames every reference covariance C_i that enters routing is block diagonal, and the diagonal projection discards the cross-block part of the moment-matched covariance. The shared affine LayerNorm (Ba et al., 2016) has $\epsilon = 10^{-6}$, and the coordinate residual MLP uses exact GELU and no dropout. The linear decoder has its own $50{,}257 \times 128$ matrix, untied from the prior means, a learned bias, and $\kappa_{\mathrm{uni}} = 1$ with the add-one-smoothed unigram frequencies of the training stream.

F.2 Initialization and optimization

The prior means start from independent normal entries of standard deviation 0.065, the prior variances at 15 through log-variance parameters equal to $\log 15$, the token-frame generators from entries of standard deviation 0.035, and the positional generators from entries of standard deviation 0.02. The comparison maps start with orthonormal rows, and the content maps at T equal to the block projector and stored $O = 0$, so $\widetilde{O}T = I$ in the exponential chart. Optimization uses fused AdamW (Loshchilov and Hutter, 2019) with $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. The base learning rates are 5×10^{-3} for the prior means, the decoder matrix and bias, the MLPs, and the normalization, 10^{-3} for the log-

variances, 3×10^{-3} for the token and positional frames, 10^{-2} for the comparison maps, and 2.5×10^{-3} for the content maps, each with 100 linear warmup steps and half-cosine decay to one percent of its base value. Decoupled weight decay is 0.1 on the token and positional frames and 0.01 on the MLPs, and no other parameter is decayed. Gradient norms are clipped to one separately for the mean, variance, and frame parameter roles. No projection is applied to the frame generators after an update. Training batches are 32 non-overlapping 128-token windows, shuffled with data seed 3, with 28,525 steps per pass. Besides the five-fold token budget, the chart, the source prior (uniform causal against ALiBi), and the covariance contract (moment matching against passthrough), the evaluated run differs from the affine realization at $K = 128$ of Table 3 in the learning rates and weight decay, the frame-generator projection, the transport chart bound, the single-precision matrix-product setting, the content initialization, and the code revision.

F.3 Numerical policies

The saved environment identifies one NVIDIA GeForce RTX 5090 and PyTorch `2.10.0.dev20251210+cu128`; training took 58,897 seconds. The forward pass runs under bfloat16 automatic mixed precision, which reaches the MLP matrix products and GELU. Routing, content aggregation, the frame products, and the moment-matched covariance run in IEEE single precision, the chart maps of Eq. (71) in double precision, and the matrix exponentials, layer normalization, decoding, and cross-entropy in single precision; deterministic algorithms are enforced, the single-precision matrix-product setting is `highest`, and TF32 is disabled for matrix products. The run has $H = B$ and $r_h = s_h = d_b$, and the temperature is computed as $\tau_h = \kappa_\beta \sqrt{d_b}$ with $\kappa_\beta = 1$, so $\tau_h = \sqrt{r_h}$. Reduced covariances are factorized without jitter and without covariance repair; failed factorizations would fall back to a double-precision square-root QR of the source factor, and the fallback was used in 0 of the 32,768 factorizations of each evaluation. The routing cost bound $[0, 8K]$ was active on 2 of 1,056,640 cost entries of eight fixed validation windows at step 10,000, on 1 at step 20,000, and on none at every later evaluation; its activity was not measured on the full splits. A fail-closed bound of 50 on the transport chart norm was never reached, all 150,000 updates were accepted, and the loss and gradient were finite at every logged step. On a sample of eight tokens the transport $U_i U_j^{-1}$ is flat to a relative holonomy residual of 4.6×10^{-15} in double precision, the per-block generator norms have median 3.37 and maximum 4.42 for token frames and 4.59 for positional frames, and the largest frame condition numbers are 8.0 and 12.2.

F.4 Parameter breakdown

The evaluated run has 227,041,490 parameters, one of which, a prior-bank decoding temperature, is frozen and unused by the linear decoder. The token-frame table holds $50{,}257 \times 4 \times 32^2 = 205{,}852{,}672$ of them (90.67%), the prior means and log-variances 6,432,896 each, and the decoder 6,483,153, so the vocabulary-indexed tables hold 225,201,617 (99.19%). The positional frames hold $128 \times 4{,}096 = 524{,}288$. The context-processing parameters number 1,315,584: 262,144 in the comparison and content maps (65,536 per layer), 1,053,184 in the MLPs, and 256 in the shared normalization. A forward pass reads 7,807,185 parameters per token: one row of each input table, one positional row, every decoder row, and

all context-processing parameters. Table lookups do not make this count comparable to dense-transformer compute (Table 9).

F.5 Diagnostics of the evaluated run

The explicit residual of the evaluated run is active. On eight fixed validation windows at the final step, the per-token ratio $\|m_i\|/\|\mu_i\|$ has median 0.74, 0.62, 0.74, and 1.05 at layers 0 to 3, and deleting μ_i from $\mu_i + m_i$ changes the normalized state by a median relative 0.81, 0.85, 0.77, and 0.55; these are forward measurements of the state, not deletion interventions on the loss. The largest singular values of the head transfers $\widetilde{O}_h T_h$ lie between 1.1 and 4.5 across heads and layers, while the comparison maps grew in common scale from 1 to between 22.5 and 26.9 along the redundancy $(S_{Q,h}, S_{K,h}) \mapsto (AS_{Q,h}, AS_{K,h})$ of Theorem 11, which leaves every cost unchanged; their scale-free relative gains lie between 0.155 and 4.70. With the trained model frozen, removing the routing costs on the first validation batch of 32 windows raises cross-entropy from 3.146 to 4.866 nats, by 1.72 nats; because the source prior is uniform, flattening it changes nothing. On eight validation windows the layer-averaged self-weights $\beta_{ii}^{(h)}$ lie between 0.025 and 0.095 and the mean attended distances between 10.8 and 20.3 positions, and one head of layer 0 attends at mean distance 1.1 with normalized entropy 0.054. The stored log-variances, which carry no weight decay, moved by 7.5% of their initial norm, while the moment-matched covariance varies with context, with median belief-covariance condition numbers 50.1, 87.2, 115.9, and 9.38 by layer. No identity-replacement probe of the covariances exists for this run. For the trained links of Section 7.4, the spectrum of each stored generator $L^{(\ell)} = O^{(\ell)} T^{(\ell)}$ was computed in double precision from the final weights, and that of the group element by spectral mapping, $\operatorname{spec} e^{L} = \exp(\operatorname{spec} L)$, since a direct eigendecomposition of the computed exponential is unreliable at these magnitudes. The traces are $\operatorname{tr} L^{(\ell)} = -2611$, -1741, -1790, and -674 for $\ell = 0, \ldots, 3$, and the smallest real parts of the eigenvalues are -66.8, -47.1, -123, and -39.5; $\|e^{L^{(\ell)}}\|_2$ lies between 1.9 and 4.2, and between 35 and 79 eigenvalues per layer have real part below -6.9, a gain below 10^{-3}. These spectra take layer 0 to be the parameter group `relational_operators` and layers 1 to 3 the groups `relational_operators_extra.0` to `.2`, the order in which the reference code indexes the layers.

F.6 Evaluation protocol

Window k of a split starts at token $32k$ and scores its last 32 targets, and the first window scores all 128, so each target is scored once and all but the first 95 targets of a split have at least 96 tokens of context. The test split gives 8,764 windows and 280,541 scored targets, and the validation split 7,654 windows and 245,020 scored targets. Validation was evaluated every 10,000 steps. The headline test evaluation uses the final weights; the test split was also evaluated at the best-validation weights, and both appear in Table 4. The fixed training sample of that table is a seeded draw of training windows scored under the evaluation protocol with about as many targets as validation. Tokenization is tiktoken 0.12.0 with the GPT-2 encoding and $V = 50{,}257$, recorded with a hash of its merge ranks; the training, validation, and test token files have 116,840,318, 245,021, and 280,542 tokens and share their hashes with the other runs reported here. No bootstrap intervals were computed for this run.

F.7 Provenance

The run directory is `34.48_wikitext-103_K128_block_glk_linear_mlp_s6`, with configuration and resolved methods in `config.json`, evaluation in `summary.json` and `test_results.json`, and source and data identity in `provenance.json`. The Git revision was `891cb1ab65d4` at launch and at the end, with a dirty working tree at both ends that included edits to the configuration module and the training driver, and the recorded code identity changed during the run. The run saved archives of the executed source at launch and at the end; recomputing their package identities reproduces the recorded launch and final code identities, and the two archives differ only in one comment line of the training driver. The executed source, including the uncommitted edits, is therefore preserved, although the Git revision alone does not identify it.

F.8 Configuration of the chart runs at width 20

Each run of Section 7.5 has $K = 20$, two heads of width 10 with block-diagonal $\mathrm{GL}(10) \times \mathrm{GL}(10)$ frames, two layers, residual coefficient $c_{\mathrm{res}} = 0$, moment-matched covariance with floor $\epsilon_Q = 10^{-4}$, local-diagonal storage, affine LayerNorm followed by a coordinate GELU MLP of expansion eight, and the linear decoder. Positions enter through rotary rotations of the routing frame with base 100, not applied to values; there is no learned positional frame, and the source prior is uniform on the causal support. Training used seed 6, the same data order in every arm, 15,000 AdamW updates of batch 64 at context 128, about 1.05 passes over the training tokens, and bfloat16 automatic mixed precision. Every arm starts from the same effective content maps, since the block-projector initialization $T = I$ with $O = I$, 0, and 0 in the affine, first-order, and exponential charts gives $\widetilde{O} = I$ in each. Evaluation follows the protocol of Section 7.2, with context 128, stride 32, and 280,541 scored test targets. Each model has 13,133,478 parameters, of which 16,401 lie outside the per-type tables and 3,200 in the comparison and content maps.

F.9 Reserved comparisons

Table 7 reserves the matched comparison of Section 7.4, Table 10 its per-seed values, Tables 8 and 9 the reproducibility and cost record of the final runs, and Table 11 the frozen interventions on the evaluated model.

Model arm	Seeds	Parameters	Test CE	Test PPL	GPU-hours
Retraining arms at the evaluated configuration and token budget					
Evaluated: learned frames, exponential chart, $c_{\text{res}} = 1$, moment-matched covariance	pending	pending	pending	pending	pending
Attention residual removed, $c_{\text{res}} = 0$	pending	pending	pending	pending	pending
Affine content chart	pending	pending	pending	pending	pending
First-order content chart	pending	pending	pending	pending	pending
Exponential chart with the channel receiver	pending	pending	pending	pending	pending
Covariance passthrough instead of moment matching	pending	pending	pending	pending	pending
Frames frozen at their random initialization	pending	pending	pending	pending	pending
Identity frames	pending	pending	pending	pending	pending
Identity frames, per-type additive table of equal size	pending	pending	pending	pending	pending
LayerNorm and MLP in reference coordinates	pending	pending	pending	pending	pending
Intertwiner-constrained nonlinear block	pending	pending	pending	pending	pending
MLP removed	pending	pending	pending	pending	pending
MLP replaced by Gaussian template routing	pending	pending	pending	pending	pending
Conventional transformers: same tokenizer, context, stride, and budget					
Matched context-processing parameters	pending	pending	pending	pending	pending
Matched measured compute	pending	pending	pending	pending	pending
Per-type block map of the frame table's size	pending	pending	pending	pending	pending
Word-level protocol, every target scored					
Gaussian attention	pending	pending	pending	pending	pending
Conventional transformer, matched	pending	pending	pending	pending	pending

Table 7: Reserved main results under one declared protocol. Each retraining arm changes one component of the evaluated realization; the two reference-coordinate and intertwiner arms make the complete stack frame-covariant. Report the mean and standard deviation over at least three replicates that vary both the model seed and the data-order seed, paired across arms, with per-seed values in Table 10. Give parameters as context-processing (non-embedding) and total counts, since per-type tables hold most of the storage. Word-level rows permit comparison with published WikiText-103 results and are not compared with the BPE rows. Cells remain pending until each evaluation is complete.

Reporting field	Final-run value or artifact identifier
Source and configuration	pending: source archive and hash, configuration schema, resolved operator selectors
Data and tokenizer	pending: dataset and split revision, tokenizer and version, vocabulary, concatenation policy, token-cache hashes, unique training tokens, passes
Architecture	pending: K, blocks, heads, ranks r_h, s_h, depth, normalization placement, residual, MLP width, decoder
Initialization	pending: prior means and covariances, frame generators, interface maps, as executed
Optimization	pending: learning rate and weight decay for every parameter table, schedule, warmup, clipping groups, post-update frame projection
Numerical realization	pending: precision per operation, cost bound, reduced-covariance factorization and failure policy, matrix-exponential policy
Training budget	pending: token presentations, updates, batch and accumulation, device-hours
Evaluation	pending: checkpoint-selection rule, context, stride, scored targets, masking

Table 8: Reproducibility contract for the final reported runs. Populate each field from the executing source archive and its saved metadata, and separate effective settings from accepted but inactive selectors.

Parameter role or resource	Evaluated run	Final run
Per-type tables (token frames, prior means and covariances, decoder)	225,201,617	pending
Positional frame generators	524,288	pending
Context-processing parameters (interface maps, MLP, normalization)	1,315,584	pending
Parameters read per token	7,807,185	pending
Optimizer state and peak device memory	1.82 GB; 15,225 MB	pending
Training throughput and device-hours	10,585 tokens/s (median); 16.36 h	pending
Inference latency at declared batch and context	pending	pending

Table 9: Storage and cost record. The evaluated-run column is taken from the saved resource record of the run of Section 7.2 on one RTX 5090 in bfloat16 mixed precision; the final-run column is reserved. Sparse access to per-type tables reduces the parameters read per token but not the stored parameters; report both, together with measured throughput under a declared precision.

Model arm	Seeds	Selected step	Validation CE	Test CE	Test PPL
pending	pending	pending	pending	pending	pending

Table 10: Reserved per-seed results for every arm of the main results table. List the model seed and the data-order seed, and apply the checkpoint-selection rule identically across arms.

Frozen intervention on every validation target	CE	ΔCE	95% interval
None (final weights)	pending	0	pending
Routing costs removed; uniform causal prior retained	pending	pending	pending
Residual term removed at every layer, $\mu_i + m_i \mapsto m_i$	pending	pending	pending
Stored covariances Σ_p^c replaced by the identity	pending	pending	pending
All routing covariances C_i replaced by the identity	pending	pending	pending
Token frames replaced by the identity in every role	pending	pending	pending
Positional frames replaced by the identity in every role	pending	pending	pending
Frames removed from the input moments only (F1)	pending	pending	pending
Frames removed from the routing covariance only (F2)	pending	pending	pending
Frames removed from the normalization and MLP conjugation only (F3)	pending	pending	pending
Frames removed from the decoder only (F4)	pending	pending	pending

Table 11: Reserved frozen interventions on the complete validation split. Each row modifies one input of the trained predictor and records the change in cross-entropy, with intervals from an article-level bootstrap. The role-wise frame rows follow Figure 3. These are properties of one trained model; they are not retraining ablations and do not establish that any component is necessary.

References

S. Amari. *Information Geometry and Its Applications*. Springer, 2016.

Shun-ichi Amari and Hiroshi Nagaoka. *Methods of Information Geometry*, volume 191 of *Translations of Mathematical Monographs*. American Mathematical Society and Oxford University Press, 2000.

Ben Athiwaratkun and Andrew Gordon Wilson. Multimodal word distributions. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1645–1656, 2017. doi: 10.18653/v1/P17-1151.

Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.

Jongseong Bae, Byung Do Cheon, and Ha Young Kim. Pro-Attention: Efficient probability distribution matching-based attention through feature space conversion. *IEEE Access*, 10:131192–131201, 2022. doi: 10.1109/ACCESS.2022.3229055.

Hareesh Bahuleyan, Lili Mou, Olga Vechtomova, and Pascal Poupart. Variational attention for sequence-to-sequence models. In *Proceedings of the 27th International Conference on Computational Linguistics*, pages 1672–1682, Santa Fe, New Mexico, USA, August 2018. Association for Computational Linguistics. URL `https://aclanthology.org/C18-1142/`.

Federico Barbero, Cristian Bodnar, Haitz Sáez de Ocáriz Borde, Michael Bronstein, Petar Veličković, and Pietro Liò. Sheaf neural networks with connection Laplacians, 2022. URL `https://arxiv.org/abs/2206.08702`. ICML 2022 Workshop on Topology, Algebra, and Geometry in Machine Learning.

Marco Baroni and Roberto Zamparelli. Nouns are vectors, adjectives are matrices: Representing adjective-noun constructions in semantic space. In *Proceedings of the 2010 Conference on Empirical Methods in Natural Language Processing*, pages 1183–1193, 2010. URL `https://aclanthology.org/D10-1115/`.

Iz Beltagy, Matthew E Peters, and Arman Cohan. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*, 2020.

Arthur S. Bianchessi, Yasmin C. Aguirre, Rodrigo C. Barros, and Lucas S. Kupssinskü. Bayesian attention mechanism: A probabilistic framework for positional encoding and context length extrapolation. In *International Conference on Learning Representations*, 2026.

D.M. Blei, A. Kucukelbir, and J.D. McAuliffe. Variational inference: a review for statisticians. *Journal of the American Statistical Association*, 112(518):859–877, 2017.

Cristian Bodnar, Francesco Di Giovanni, Benjamin Paul Chamberlain, Pietro Liò, and Michael M. Bronstein. Neural sheaf diffusion: A topological perspective on heterophily and oversmoothing in GNNs. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.

Denis Boyda, Gurtej Kanwar, Sébastien Racanière, Danilo Jimenez Rezende, Michael S. Albergo, Kyle Cranmer, Daniel C. Hackett, and Phiala E. Shanahan. Sampling using $SU(N)$ gauge equivariant flows. *Physical Review D*, 103:074504, 2021. doi: 10.1103/PhysRevD.103.074504.

Michael M. Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478*, 2021.

Taco S. Cohen, Maurice Weiler, Berkay Kicanaoglu, and Max Welling. Gauge equivariant convolutional networks and the icosahedral CNN. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR, 2019. URL https://proceedings.mlr.press/v97/cohen19d.html.

M. Creutz. *Quarks, Gluons and Lattices*. Cambridge University Press, Cambridge, 1983. ISBN 978-0521244053.

Walter J. Culver. On the existence and uniqueness of the real logarithm of a matrix. *Proceedings of the American Mathematical Society*, 17(5):1146–1151, 1966.

Zihang Dai, Zhilin Yang, Yiming Yang, Jaime Carbonell, Quoc V Le, and Ruslan Salakhutdinov. Transformer-XL: Attentive language models beyond a fixed-length context. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2978–2988, 2019.

Abhishek Das, Théophile Gervet, Joshua Romoff, Dhruv Batra, Devi Parikh, Michael Rabbat, and Joelle Pineau. TarMAC: Targeted multi-agent communication. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, 2019.

Yuntian Deng, Yoon Kim, Justin Chiu, Demi Guo, and Alexander M. Rush. Latent alignment and variational attention. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/hash/b691334ccf10d4ab144d672f7783c8a3-Abstract.html.

Michael R. Douglas and Gregory Moore. D-branes, quivers, and ALE instantons, 1996. URL https://arxiv.org/abs/hep-th/9603167.

Simon S. Du, Wei Hu, and Jason D. Lee. Algorithmic regularization in learning deep homogeneous models: Layers are automatically balanced. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Tohru Eguchi and Hikaru Kawai. Reduction of dynamical degrees of freedom in the large-N gauge theory. *Physical Review Letters*, 48(16):1063–1066, 1982. doi: 10.1103/PhysRevLett.48.1063.

Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021.

Xinjie Fan, Shujian Zhang, Bo Chen, and Mingyuan Zhou. Bayesian attention modules. In *Advances in Neural Information Processing Systems*, volume 33, 2020. URL `https://arxiv.org/abs/2010.10604`.

Matteo Favoni, Andreas Ipp, David I. Müller, and Daniel Schuh. Lattice gauge equivariant convolutional neural networks. *Physical Review Letters*, 128(3):032003, 2022. doi: 10.1103/PhysRevLett.128.032003.

Jakob N. Foerster, Yannis M. Assael, Nando de Freitas, and Shimon Whiteson. Learning to communicate with deep multi-agent reinforcement learning. *Advances in Neural Information Processing Systems*, 29, 2016.

Cédric Fournel, Jordan François, Serge Lazzarini, and Thierry Masson. Gauge invariant composite fields out of connections, with examples. *International Journal of Geometric Methods in Modern Physics*, 11:1450016, 2014. doi: 10.1142/S0219887814500169.

Jordan François. Artificial versus substantial gauge symmetries: A criterion and an application to the electroweak model. *Philosophy of Science*, 86(3):472–496, 2019. doi: 10.1086/703571.

Theodore Frankel. *The Geometry of Physics: An Introduction*. Cambridge University Press, 3rd edition, 2011.

Fabian B. Fuchs, Daniel E. Worrall, Volker Fischer, and Max Welling. SE(3)-transformers: 3D roto-translation equivariant attention networks. In *Advances in Neural Information Processing Systems*, volume 33, pages 1970–1981, 2020.

Prasad Gabbur, Manjot Bilkhu, and Javier R. Movellan. Probabilistic attention for interactive segmentation. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

Borjan Geshkovski, Cyril Letrouit, Yury Polyanskiy, and Philippe Rigollet. A mathematical perspective on transformers. *arXiv preprint*, 2023.

Justin Gilmer, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. Neural message passing for quantum chemistry. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1263–1272, 2017. URL `https://proceedings.mlr.press/v70/gilmer17a.html`.

David Ha, Andrew M. Dai, and Quoc V. Le. Hypernetworks. In *International Conference on Learning Representations*, 2017. URL `https://arxiv.org/abs/1609.09106`.

B.C. Hall. *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*. Springer, 2nd edition, 2015.

Jakob Hansen and Thomas Gebhart. Sheaf neural networks, 2020. URL https://arxiv.org/abs/2012.06333. NeurIPS 2020 Workshop on Topological Data Analysis and Beyond.

Jakob Hansen and Robert Ghrist. Toward a spectral theory of cellular sheaves. *Journal of Applied and Computational Topology*, 3:315–358, 2019. doi: 10.1007/s41468-019-00038-7.

Jakob Hansen and Robert Ghrist. Opinion dynamics on discourse sheaves. *SIAM J. Appl. Math.*, 81(5):2033–2060, 2021. doi: 10.1137/20M1341088.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016. doi: 10.1109/CVPR.2016.90.

Lingshen He, Yiming Dong, Yisen Wang, Dacheng Tao, and Zhouchen Lin. Gauge equivariant transformer. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://proceedings.neurips.cc/paper/2021/hash/e57c6b956a6521b28495f2886ca0977a-Abstract.html.

Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (GELUs). *arXiv preprint arXiv:1606.08415*, 2016.

N.J. Higham. *Functions of Matrices: Theory and Computation*. SIAM, Philadelphia, 2008.

Marlis Hochbruck and Alexander Ostermann. Exponential integrators. *Acta Numerica*, 19: 209–286, 2010. doi: 10.1017/S0962492910000048.

Benjamin Hoover, Yuchen Liang, Bao Pham, Rameswar Panda, Hendrik Strobelt, Duen Horng Chau, Mohammed J. Zaki, and Dmitry Krotov. Energy transformer. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/57a9b97477b67936298489e3c1417b0a-Abstract-Conference.html.

Siddhant M. Jayakumar, Wojciech M. Czarnecki, Jacob Menick, Jonathan Schwarz, Jack Rae, Simon Osindero, Yee Whye Teh, Tim Harley, and Razvan Pascanu. Multiplicative interactions and where to find them. In *International Conference on Learning Representations*, 2020. URL https://mlanthology.org/iclr/2020/jayakumar2020iclr-multiplicative/.

Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7B. *arXiv preprint arXiv:2310.06825*, 2023.

Jiechuan Jiang and Zongqing Lu. Learning attentional communication for multi-agent cooperation. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL https://arxiv.org/abs/1805.07733.

Chaitanya K. Joshi. Transformers are graph neural networks, 2025. URL https://arxiv.org/abs/2506.22084.

Gurtej Kanwar, Michael S. Albergo, Denis Boyda, Kyle Cranmer, Daniel C. Hackett, Sébastien Racanière, Danilo Jimenez Rezende, and Phiala E. Shanahan. Equivariant flow-based sampling for lattice gauge theory. *Physical Review Letters*, 125(12):121601, 2020. doi: 10.1103/PhysRevLett.125.121601.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020. URL `https://arxiv.org/abs/2001.08361`.

Hilbert J. Kappen, Vicenç Gómez, and Manfred Opper. Optimal control as a graphical model inference problem. *Machine Learning*, 87:159–182, 2012. doi: 10.1007/s10994-012-5278-7.

Hyunjik Kim, George Papamakarios, and Andriy Mnih. The Lipschitz constant of self-attention. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 5562–5571, 2021.

Yoon Kim, Carl Denton, Luong Hoang, and Alexander M. Rush. Structured attention networks. In *International Conference on Learning Representations*, 2017. URL `https://openreview.net/forum?id=HkE0Nvqlg`.

Alastair D. King. Moduli of representations of finite dimensional algebras. *The Quarterly Journal of Mathematics*, 45(4):515–530, 1994. doi: 10.1093/qmath/45.4.515.

J. Kogut and L. Susskind. Hamiltonian formulation of Wilson's lattice gauge theories. *Physical Review D*, 11(2):395–408, 1975. doi: 10.1103/physrevd.11.395.

John B. Kogut. An introduction to lattice gauge theory and spin systems. *Reviews of Modern Physics*, 51(4):659–713, 1979. doi: 10.1103/RevModPhys.51.659.

Daniel Kunin, Javier Sagastuy-Brena, Surya Ganguli, Daniel L. K. Yamins, and Hidenori Tanaka. Neural mechanics: Symmetry and broken conservation laws in deep learning dynamics. In *International Conference on Learning Representations*, 2021.

Angeliki Lazaridou and Marco Baroni. Emergent multi-agent communication in the deep learning era, 2020. URL `https://arxiv.org/abs/2006.02419`.

Angeliki Lazaridou, Alexander Peysakhovich, and Marco Baroni. Multi-agent cooperation and the emergence of (natural) language. In *International Conference on Learning Representations*, 2017. URL `https://arxiv.org/abs/1612.07182`.

Sergey Levine. Reinforcement learning and control as probabilistic inference: Tutorial and review. arXiv preprint arXiv:1805.00909, 2018.

Elon Litman. Scaled-dot-product attention as one-sided entropic optimal transport. arXiv preprint arXiv:2508.08369, 2025.

Elon Litman and Gabe Guo. You need better attention priors. arXiv preprint arXiv:2601.15380, 2026.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.

Florian Mai, Lukas Galke, and Ansgar Scherp. CBOW is not all you need: Combining CBOW with the compositional matrix space model. In *International Conference on Learning Representations*, 2019. URL `https://arxiv.org/abs/1902.06423`.

André F. T. Martins and Ramón Fernandez Astudillo. From softmax to sparsemax: A sparse model of attention and multi-label classification. In *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1614–1623, 2016. URL `https://proceedings.mlr.press/v48/martins16.html`.

Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. *arXiv preprint arXiv:1609.07843*, 2017.

Melissa B. Miller and Bonnie L. Bassler. Quorum sensing in bacteria. *Annual Review of Microbiology*, 55:165–199, 2001. doi: 10.1146/annurev.micro.55.1.165.

Beren Millidge, Alexander Tschantz, and Christopher L. Buckley. Predictive coding approximates backprop along arbitrary computation graphs. *Neural Computation*, 34(6): 1329–1368, 2022. doi: 10.1162/neco_a_01497.

Takeru Miyato, Bernhard Jaeger, Max Welling, and Andreas Geiger. GTA: A geometry-aware attention mechanism for multi-view transformers. In *International Conference on Learning Representations*, 2024. URL `https://arxiv.org/abs/2310.10375`.

Igor Mordatch and Pieter Abbeel. Emergence of grounded compositional language in multi-agent populations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018. doi: 10.1609/aaai.v32i1.11492.

Javier R. Movellan and Prasad Gabbur. Probabilistic transformers. arXiv preprint arXiv:2010.15583, 2020.

Toshiyuki Nakagaki, Hiroyasu Yamada, and Ágota Tóth. Maze-solving by an amoeboid organism. *Nature*, 407(6803):470, 2000. doi: 10.1038/35035159.

Mikio Nakahara. *Geometry, topology and physics*. Institute of Physics Publishing, Bristol, 2nd edition, 2003.

Tam Nguyen, Tan M. Nguyen, Dung D. Le, Duy Khuong Nguyen, Viet-Anh Tran, Richard G. Baraniuk, Nhat Ho, and Stanley J. Osher. Improving transformers with probabilistic attention keys. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 16595–16621, 2022.

Vlad Niculae and Mathieu Blondel. A regularized framework for sparse and structured neural attention. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Sophie Ostmeier, Brian Axelrod, Maya Varma, Michael E. Moseley, Akshay Chaudhari, and Curtis Langlotz. LieRE: Lie rotational positional encodings. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025. URL https://arxiv.org/abs/2406.10322.

Ofir Press, Noah A Smith, and Mike Lewis. Train short, test long: Attention with linear biases enables input length extrapolation. In *International Conference on Learning Representations*, 2022.

Lulu Qian, Erik Winfree, and Jehoshua Bruck. Neural network computation with DNA strand displacement cascades. *Nature*, 475(7356):368–372, 2011. doi: 10.1038/nature10262.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. *OpenAI Blog*, 1(8):9, 2019.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140): 1–67, 2020.

Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Thomas Adler, Lukas Gruber, Markus Holzleitner, Milena Pavlović, Geir Kjetil Sandve, et al. Hopfield networks is all you need. In *International Conference on Learning Representations*, 2021.

Sebastian Rudolph and Eugenie Giesbrecht. Compositional matrix-space models of language. In *Proceedings of the 48th Annual Meeting of the Association for Computational Linguistics*, pages 907–916, 2010. URL https://aclanthology.org/P10-1093/.

Amit Singer and Hau-Tieng Wu. Vector diffusion maps and the connection Laplacian. *Commun. Pure Appl. Math.*, 65(8):1067–1144, 2012. doi: 10.1002/cpa.21395.

L. T. Skovgaard. A Riemannian geometry of the multivariate normal model. *Scandinavian Journal of Statistics*, 11(4):211–223, 1984.

Richard Socher, Brody Huval, Christopher D. Manning, and Andrew Y. Ng. Semantic compositionality through recursive matrix-vector spaces. In *Proceedings of the 2012 Joint Conference on Empirical Methods in Natural Language Processing and Computational Natural Language Learning*, pages 1201–1211, 2012. URL https://aclanthology.org/D12-1110/.

David Soloveichik, Matthew Cook, Erik Winfree, and Jehoshua Bruck. Computation with finite stochastic chemical reaction networks. *Natural Computing*, 7(4):615–633, 2008. doi: 10.1007/s11047-008-9067-y.

Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.

Sainbayar Sukhbaatar, Arthur Szlam, and Rob Fergus. Learning multiagent communication with backpropagation. In *Advances in Neural Information Processing Systems*, volume 29, 2016. URL `https://arxiv.org/abs/1605.07736`.

Atsushi Tero, Seiji Takagi, Tetsu Saigusa, Kentaro Ito, Dan P. Bebber, Mark D. Fricker, Kenji Yumiki, Ryo Kobayashi, and Toshiyuki Nakagaki. Rules for biologically inspired adaptive network design. *Science*, 327(5964):439–442, 2010. doi: 10.1126/science.1177894.

Marc Toussaint and Amos Storkey. Probabilistic inference for solving discrete and continuous state Markov decision processes. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 945–952, 2006. doi: 10.1145/1143844.1143963.

Yao-Hung Hubert Tsai, Shaojie Bai, Makoto Yamada, Louis-Philippe Morency, and Ruslan Salakhutdinov. Transformer dissection: An unified understanding for transformer's attention via the lens of kernel. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 4343–4352, 2019.

Tim van Erven and Peter Harremøes. Rényi divergence and Kullback–Leibler divergence. *IEEE Transactions on Information Theory*, 60(7):3797–3820, 2014. doi: 10.1109/TIT.2014.2320500.

Charles F. Van Loan. Computing integrals involving the matrix exponential. *IEEE Transactions on Automatic Control*, 23(3):395–404, 1978. doi: 10.1109/TAC.1978.1101743.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, pages 5998–6008, 2017.

Luke Vilnis and Andrew McCallum. Word representations via Gaussian embedding. In *International Conference on Learning Representations*, 2015.

M.J. Wainwright and M.I. Jordan. Graphical models, exponential families, and variational inference. *Foundations and Trends in Machine Learning*, 1(1–2):1–305, 2008. doi: 10.1561/2200000001.

Maurice Weiler, Patrick Forré, Erik Verlinde, and Max Welling. Coordinate independent convolutional networks: Isometry and gauge equivariant convolutions on Riemannian manifolds, 2021.

Christof Wetterich. Linear lattice gauge theory. *Nuclear Physics B*, 884:44–65, 2014. doi: 10.1016/j.nuclphysb.2014.04.002.

James C. R. Whittington and Rafal Bogacz. An approximation of the error backpropagation algorithm in a predictive coding network with local Hebbian synaptic plasticity. *Neural Computation*, 29(5):1229–1262, 2017. doi: 10.1162/NECO_a_00949.

K.G. Wilson. Confinement of quarks. *Physical Review D*, 10(8):2445–2459, 1974. doi: 10.1103/physrevd.10.2445.